Qualcomm Technologies, Inc.

Device description

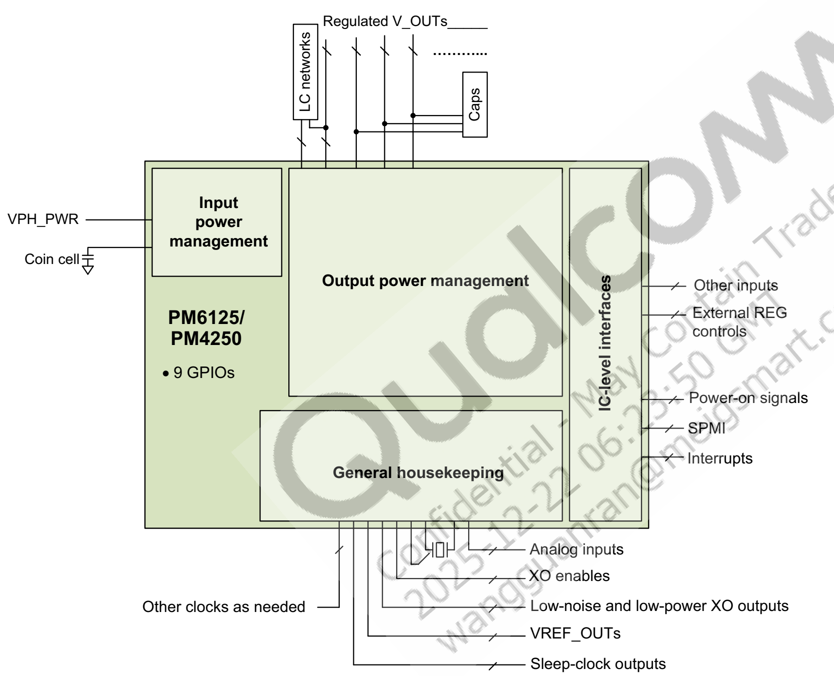
The PM6125/PM4250 device integrates all the wireless product’s power management, general housekeeping, and IC-level interface support functions into a single mixed-signal IC.

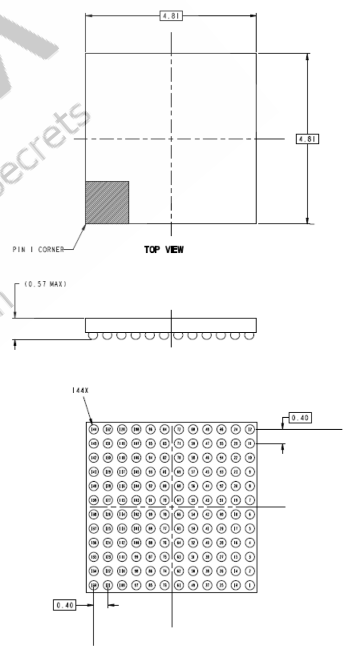
- PM6125 is optimized for the SM6125 chipset
- PM4250 is optimized for the SM4250/SM6115 chipset
- System-clock and sleep-clock sources for entire chipset
  - Two RF (low-noise) outputs
  - Three baseband (low-noise) output
  - Sleep clock output

Key features (see Section 1.2 for details)

- Three HFS510 SMPS and five FTS510 SMPS
- 24 LDO linear regulators
- On-chip ADC
- Overtemperature protection
- 38.4 MHz XO controller and XO outputs
- Sleep clock and RTC with alarm
- SPMI
- 9 GPIOs
- 144-pin fan-out wafer-level nanoscale package (144 FOWNSP)

PM6125/PM4250 high-level block diagram and 144 FOWNSP package outline drawing





Contents

1 Introduction

Document updates

See the Revision history for details on the changes included in this revision.

1.1 Functional block diagram

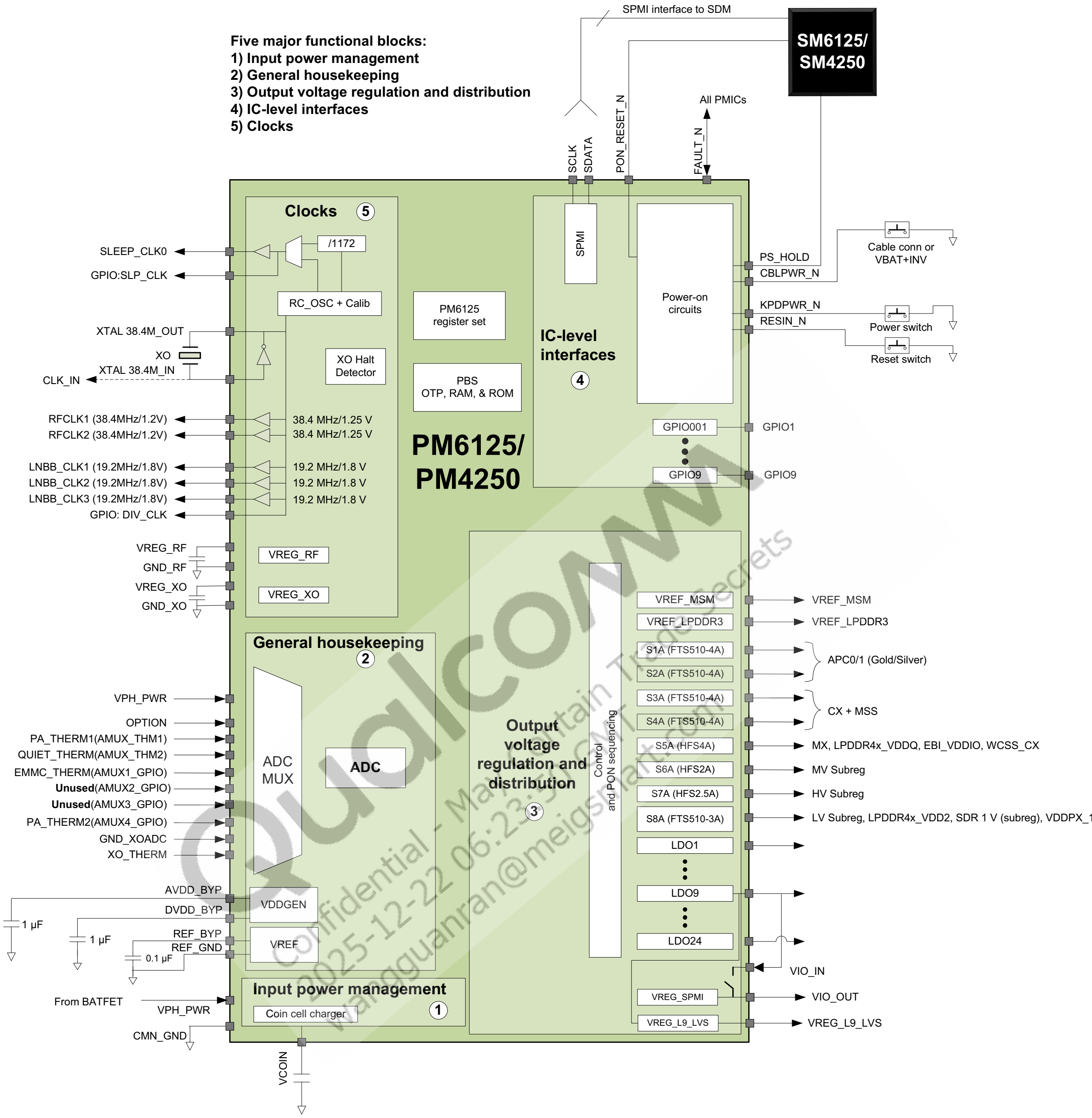


Figure 1-1 PM6125/PM4250 functional block diagram and example application

1.2 PM6125/PM4250 features

NOTE: Some of the hardware features integrated within the PM6125/PM4250 must be enabled by the modem IC software. See the latest revision of the applicable software release notes to identify the enabled PM6125/PM4250 features.

Table 1-1 PM6125/PM4250 features

Feature	PM6125/PM4250 capability
Input power management	
Coin cell or capacitor backup	Keep-alive power source; orchestrated charging
Output voltage regulation	
Switched-mode power supplies HF-SMPS FT-SMPS	 Three Five
Low-dropout linear regulators	24
Pseudo-capless LDO designs	15
General housekeeping	
On-chip ADC	Shared housekeeping (HK) and XO support
Analog multiplexing for ADC HK inputs XO input	 Many internal nodes and external inputs Dedicated pin (XO_THERM)
Overtemperature protection	Multistage smart thermal control
38.4 MHz oscillator support	XO (with on-chip ADC)
XO controller and XO outputs	Five sets: three low-noise baseband outputs and two low-power RF outputs
Special purpose clock outputs	Sleep clock; 19.2, 9.6, 4.8, 2.4, and 1.2 MHz, including low-power mode 2.4 MHz for MP3; three high-speed GPIOs for fast clocks
Real-time clock (RTC)	RTC clock circuits and alarms
IC-level interfaces	
Primary status and control	Two-line SPMI
Interrupt managers	Supported by SPMI
Power sequencing	Power-on, power-off, and soft resets
Configurable I/Os	
GPIO pins	9; configurable as digital inputs or outputs or analog pass-through
Package	
Size	4.81 × 4.81 × 0.57 mm
Pin count and package type	FOWNSP144

2 Pin definitions

2.1 I/O parameter definitions

Table 2-1 I/O parameter (pad type) definitions

Symbol	Description
Pad attribute	
AI	Analog input
AO	Analog output
DI	Digital input (CMOS)
DO	Digital output (CMOS)
PI	Power input; a pin that handles 10 mA or more of current flow into the device
PO	Power output; a pin that handles 10 mA or more of current flow out of the device
Z	High-impedance (Hi-Z) output
MV	Medium voltage
LV	Low voltage
GNDP	Power ground; a pad that handles 10 mA or more of current flow returning to ground. Layout considerations must be made for these pads.
GNDC	Common ground; a pad that does not handle a significant amount of current flow, typically used for grounding digital circuits and substrates
Pad voltage groupings	
V_INT	Internally generated supply voltage for some power-on circuits
V_PAD	Supply for modem IC interfaces; connected internally to VDD_MSM_IO
V_XBB	Supply for LN_BB_CLKx output buffer; connected internally to VREG_BB_CLK
V_XRF	Supply for RF_CLKx output buffers; connected internally to VREG_RF_CLK
GPIO pin configurations	
When configured as inputs, GPIO pins have configurable pull settings.	
NP	No internal pull enabled
PU	Internal pull-up enabled
PD	Internal pull-down enabled
When configured as outputs, GPIO pins have configurable drive strengths that depend on the GPIO pad's supply voltage. See Chapter 3 for details.	

2.2 Pin assignments

The PM6125/PM4250 is available in the 144FOWNSP. See Chapter 4 for package details.

A high-level view of the pin assignments is shown in Figure 2-1.

Figure 2-1 PM6125/PM4250 pin assignments (bottom view)

2.2.1 Pin descriptions

Descriptions of all pins are presented in the following tables, organized by functional group:

Table 2-2 Pin descriptions – input power management functions

Pad #	Pad name and/or function	Pad type [1]	Functional description
Input power sources			
102	VPH_PWR	PI	System input power node generated by either the charger or battery
Coin cell or keep-alive battery			
135	VCOIN	PI, PO	Coin-cell charge and supply

1. See Table 2-1 for pad voltage and type definitions.

Table 2-3 Pin descriptions – output power management functions

Pad #	Pad name and/or function	Pad type [1]	Functional description
High-frequency buck SMPS circuits			
12, 24, 35	VSW_S5	PO	S5 SMPS switch node
34	VREG_S5	AI	S5 SMPS sense input
36	VDD_S5	PI	S5 SMPS supply power input
11, 23	GND_S5	GNDP	Switching ground for VREG_S5
52	VREG_S6	AI	S6 SMPS sense input
25	VDD_S6	PI	S6 SMPS supply power input
1, 14	GND_S6	GNDP	Switching ground for VREG_S6
13, 26	VSW_S6	PO	S6 SMPS switch node
7, 19	VSW_S7	PO	S7 SMPS switch node
54	VREG_S7	AI	S7 SMPS sense input
8	VDD_S7	PI	S7 SMPS supply power input

Table 2-3 Pin descriptions – output power management functions (cont.)

Pad #	Pad name and/or function	Pad type [1]	Functional description
6	GND_S7	GNDP	Switching ground for VREG_S7
Fast-transient buck SMPS circuits			
59, 60	VSW_S1	PO	S1 SMPS switch node
58	VREG_S1	AI	S1 feedback node
46	RMT_GND_S1	GNDP	Remote ground sense at the MSM device
48	VDD_S1	PI	S1 SMPS supply power input
71, 72	GND_S1	GNDP	Switching ground for VREG_S1
95, 96	VSW_S2	PO	S2 SMPS switch node
70	VREG_S2	AI	S2 SMPS sense input
82	RMT_GND_S2	GNDP	Remote ground sense at the MSM device
108	VDD_S2	PI	S2 SMPS supply power input
83, 84	GND_S2	GNDP	Switching ground for VREG_S2
119, 132, 144	VSW_S3	PO	S3 SMPS switch node
142	VREG_S3	AI	S3 SMPS sense input
130	RMT_GND_S3	GNDP	Remote ground sense at the MSM device
120	VDD_S3	PI	S3 SMPS supply power input
131, 143	GND_S3	GNDP	Switching ground for VREG_S3
128, 140	VSW_S4	PO	S4 SMPS switch node
118	VREG_S4	AI	S4 SMPS sense input
139	VDD_S4	PI	S4 SMPS supply power input
129, 141	GND_S4	GNDP	S4 SMPS power ground
117	RMT_GND_S4	GNDP	Remote ground sense at the MSM device
5, 17	GND_S8	GNDP	Switching ground for VREG_S8
4, 16	VSW_S8	PO	S8 SMPS switch node
28	VREG_S8	AI	S8 SMPS sense input
29	RMT_GND_S8	GNDP	Remote ground sense at the MSM device
3	VDD_S8	PI	S8 SMPS supply power input
Low dropout regulator (LDO) circuits			
61	VREG_L1	PO	L1 LDO regulated output
63	VREG_L2	PO	L2 LDO regulated output
64	VREG_L3	PO	L3 LDO regulated output
39	VREG_L4	PO	L4 LDO regulated output
113	VREG_L5	PO	L5 LDO regulated output
9	VREG_L6	PO	L6 LDO regulated output
49	VREG_L7	PO	L7 LDO regulated output

Table 2-3 Pin descriptions – output power management functions (cont.)

Pad #	Pad name and/or function	Pad type [1]	Functional description
33	VREG_L8	PO	L8 LDO regulated output
43	VREG_L9	PO	L9 LDO regulated output
81	VREG_L10	PO	L10 LDO regulated output
45	VREG_L11	PO	L11 LDO regulated output
69	VREG_L12	PO	L12 LDO regulated output
105	VREG_L13	PO	L13 LDO regulated output
93, 94	VREG_L14	PO	L14 LDO regulated output
114	VREG_L15	PO	L15 LDO regulated output
57	VREG_L16	PO	L16 LDO regulated output
38	VREG_L17	PO	L17 LDO regulated output
37	VREG_L18	PO	L18 LDO regulated output
126	VREG_L19	PO	L19 LDO regulated output
32	VREG_L9_LVS	PO	L9 switch output for SDR1.8 V rail
138	VREG_L20	PO	L20 LDO regulated output
136	VREG_L21	PO	L21 LDO regulated output
137	VREG_L22	PO	L22 LDO regulated output
41	VREG_L23	PO	L23 LDO regulated output
31	VREG_L24	PO	L24 LDO regulated output
50	VDD_L1_L7_L17_L18	PI	Subregulation LDO input
27, 40	VDD_L2_L3_L4	PI	Subregulation LDO input
125	VDD_L5_L15_L19_L20_L21_L22	PI	LDO input, powered from VPH_PWR or VBOB
20	VDD_L6_L8	PI	Subregulation LDO input
56	VDD_L9_L11	PI	Subregulation LDO input
68	VDD_L12_L16	PI	Subregulation LDO input
80	VDD_L10_L13_L14	PI	Subregulation LDO input
42	VDD_L23_L24	PI	LDO input, powered from VPH_PWR or VBOB

1. See Table 2-1 for pad voltage and type definitions.

Table 2-4 Pin descriptions – general housekeeping functions

Pad #	Pad name and/or function	Pad characteristics [1]		Functional description
		Voltage	Type	
Analog multiplexer and HK/XO ADC circuits				
86	AMUX_1	–	AI	Analog multiplexer (AMUX) input 1; RF PA thermistor
98	AMUX_2	–	AI	Analog multiplexer (AMUX) input 2; Quiet therm

Table 2-4 Pin descriptions – general housekeeping functions (cont.)

Pad #	Pad name and/or function	Pad characteristics [1] Voltage	Type	Functional description
85	GND_XOADC	–	GNDP	XO ADC reference ground
97	XO_THERM	–	AI	XO thermistor resistor divider input; option pin
111	VDD_LDO_XO_RF	–	PI	Clock buffer LDO input, powered from L9 on PCB
124	OPTION	–	AI	Option hardware configuration control
PMIC power infrastructure				
73	AVDD_BYP	–	PO	Internal aVdd regulator (1.8 V); with a 1 μF capacitor
75	REF_BYP	–	AO	Bandgap reference
76	REF_GND	–	GNDP	Bandgap reference ground
74	DVDD_BYP	–	PO	Internal dVdd regulator (1.5 V); with a 1 μF capacitor
Crystal oscillator (XO) circuits and buffered outputs				
133, 134	XTAL_IN	–	AI	19.2 MHz/38.4 MHz oscillator/TCXO clock input
121	XTAL_OUT	–	AO	19.2 MHz/38.4 MHz oscillator output
122	GND_XO	–	GNDP	Ground for 38.4 MHz XO
123	VREG_XO	–	PO	XO VREG (internal use only)
109	VREG_RF_CLK	–	PO	RF clock output VREG (internal use only)
101	VREG_BB	–	PO	BB_CLK_DRV VDD (connected to L10 output on PCB)
112	LN_BB_CLK1	–	DO	CXO input to the SDM device
89	LN_BB_CLK2	–	DO	WGR clock
99	LN_BB_CLK3	–	DO	NFC clock
100	RF_CLK1	–	DO	SDR clock
88	RF_CLK2	–	DO	WCN clock
110	GND_RF	–	GNDP	Ground for 19.2 MHz clock buffers
Sleep clock				
55	SLEEP_CLK	–	DO	32 kHz sleep-clock output
Vref outputs				
62	VREF_MSM	–	AO	1.25 V reference for HV PADC
51	VREF_LPDDR3	–	AO	1.25 V reference for LPDDR3. The voltage on this pin is half of the supply voltage on VDD_L2_L3_L4.
GPIOs can be configured for general housekeeping functions not listed here. [2]				

1. See Table 2-1 for pad voltage and type definitions.
2. Other housekeeping GPIO functions are possible. To assign a GPIO a particular function, identify all of your application's requirements and map each GPIO to its function, carefully avoiding assignment conflicts. All GPIOs are listed in Table 2-6.

Table 2-5 Pin descriptions – IC-level interface functions

Pad #	Pad name and/or function	Pad characteristics [1]		Functional description
		Voltage	Type	
Power-on/power-off/reset control				
107	FAULT_N	V_PAD	DI, DO	PMIC fault signal (bidirectional) that initiates shutdown or S3 reset to all PMICs
77	VIO_IN	–	PI	I/O 1.8 V supply (connected to L9 output on PCB)
87	VIO_OUT	–	PO	VREG_SPMI output for SPMI_PAD and PX0 (during boot) with the PMIC controller
115	KPD_PWR_N	–	DI	Power-on key ground switch (200 kΩ internal PU)
103	RESIN_N	–	DI	Reset input to PM (40 kΩ internal PU)
127	PS_HOLD	–	DI	Power supply hold
116	CBL_PWR_N	–	DI	Cable power-on ground switch (200 kΩ internal PU)
92	PON_RESET_N	–	DO	Reset output to SDM device's RESIN_N
System power management interface (SPMI) signals				
104	SPMI_CLK	V_PAD	DI	SPMI communication bus clock signal
91	SPMI_DATA	V_PAD	DI, DO	SPMI communication bus data signal
Miscellaneous IC functions				
2	TEST_EN_VPP	–	GNDC	Pin must be grounded externally
GPIOs can be configured for IC-level interface functions not listed here. [2]				

1. See Table 2-1 for pad voltage and type definitions.
2. Other IC-level interface GPIO functions are possible. To assign a GPIO a particular function, identify all of your application's requirements and map each GPIO to its function, carefully avoiding assignment conflicts. All GPIOs are listed in Table 2-6.

Table 2-6 Pin descriptions – general-purpose input/output functions for PM6125 on SM6125 platform

Pad #	Pad name	Configurable function	Pad type [1] [2]	Functional description
Predefined GPIO functions – available only at the assigned GPIOs				
44	GPIO_01	DIV_CLK	LV	Configurable; default digital input with 10 µA pull-down Codec MCLK 9.6 MHz
30	GPIO_02	AOSS sleep indicator	LV	Configurable; default digital input with 10 µA pull-down AOSS sleep indicator
15	GPIO_03	CAM/FLASH_THERM	LV	Configurable; default digital input with 10 µA pull-down CAM/FLASH_THERM
47	GPIO_04	BOOST_BYP_MODE_CTL	LV	Configurable; default digital input with 10 µA pull-down BOOST_BYP_MODE_CTL
21	GPIO_05	Volume + Key/ SLEEP_CLK2	LV	Configurable; default digital input with 10 µA pull-down Volume plus key

Table 2-6 Pin descriptions – general-purpose input/output functions for PM6125 on SM6125 platform (cont.)

Pad #	Pad name	Configurable function	Pad type [1] [2]	Functional description
106	GPIO_06		MV	Configurable; default digital input with 10 µA pull-down Unused
10	GPIO_07	PA_THERM1	MV	Configurable; default digital input with 10 µA pull-down RF PA_THERM1
22	GPIO_08	PWM_OUT	MV	Configurable; default digital input with 10 µA pull-down PWM_OUT for backlight
18	GPIO_09		MV	Configurable; default digital input with 10 µA pull-down Unused

1. See Table 2-1 for pad voltage and type definitions.
2. The instantaneous voltage of an MV GPIO should not exceed VPH_PWR.

Table 2-7 Pin descriptions – general-purpose input/output functions for PM4250 on SM4250/SM6115 platform

Pad #	Pad name	Configurable function	Pad type [1] [2]	Functional description
Predefined GPIO functions – available only at the assigned GPIOs				
44	GPIO_01		LV	Reserved for eLDO
30	GPIO_02	SLEEP_CLK_IN	LV	Configurable; default digital input with 10 µA pull-down SLEEP_CLK_IN (optional)
15	GPIO_03	CAM/FLASH_THERM	LV	Configurable; default digital input with 10 µA pull-down CAM/FLASH_THERM
47	GPIO_04	EMMC_UFS_THERM	LV	Configurable; default digital input with 10 µA pull-down EMMC_UFS_THERM
21	GPIO_05	Volume + Key/ SLEEP_CLK2	LV	Configurable; default digital input with 10 µA pull-down Volume plus key or additional SLEEP_CLK output
106	GPIO_06		MV	Configurable; default digital input with 10 µA pull-down Unused
10	GPIO_07	PA_THERM1	MV	Configurable; default digital input with 10 µA pull-down RF PA_THERM1
22	GPIO_08	PWM_OUT	MV	Configurable; default digital input with 10 µA pull-down PWM_OUT for backlight
18	GPIO_09		MV	Configurable; default digital input with 10 µA pull-down Unused

1. See Table 2-1 for pad voltage and type definitions.
2. The instantaneous voltage of an MV GPIO should not exceed VPH_PWR.

Table 2-8 Pin descriptions – common grounds

Pad #	Pad name	Pad type [1]	Functional description
Note: This table only includes common ground pins. Power ground pins are grouped with their respective modules, and can be found in the previous tables.			
53, 65, 66, 67, 78, 79, 90	CMN_GND	GNDC	Ground for all non-specialized circuits

1. See Table 2-1 for pad voltage and type definitions.

3 Electrical specifications

3.1 Absolute maximum ratings

The absolute maximum ratings (Table 3-1) reflect the stress levels that, if exceeded, may cause permanent damage to the device. No functionality is guaranteed outside the operating specifications. Functionality and reliability are only guaranteed within the operating conditions described in Section 3.2.

Table 3-1 Absolute maximum ratings

Parameter		Min	Max	Units
Power supply pads				
VPH_PWR	Handset power-supply voltage	-0.5	+6	V
VDD_xxx	PMIC power-supply voltages not listed elsewhere (steady state) Transient (< 10 ms)	-0.5 -0.5	+6 +7	V V
GPIO_xx	MV LV	-0.5 -0.5	6 2.2	V V
T_S	Storage temperature [1]	-40	+150	°C
ESD protection and thermal conditions – see Section 7.1, Section 7.2, and Section 4.5.				

1. The storage temperature range applies when the device is in Off mode.

3.2 Operating conditions

Operating conditions include design team-controlled parameters such as power supply voltage and thermal conditions (Table 3-2). The PM6125/PM4250 meets all performance specifications listed in Section 3.3 through Section 3.7, when used within the operating conditions, unless otherwise noted in those sections (provided the absolute maximum ratings have never been exceeded).

Table 3-2 Operating conditions

Parameter		Min	Typ	Max	Units
Power supply pads					
VPH_PWR	Handset power-supply voltage (steady state) Short-duration VPH_PWR excursions beyond normal operating voltage range (< 1 ms)	+3.2 [1] +2.7 [1]	+3.8 +3.8	+4.75 +5.25	V V
VDD_IO	Pad voltage for digital I/Os to/from the IC	+1.75	–	+1.85	V
VCOIN	Coin-cell voltage	2.0	+3.0	3.25	V

Table 3-2 Operating conditions (cont.)

Parameter		Min	Typ	Max	Units
GPIO_xx	MV LV	– –	VPH 1.8	– –	V V
Signal pins					
V_IN	Voltage on any nonpower-supply pin [2]	0	–	V_{XX} + 0.5	V
Thermal conditions					
T_A	Ambient operating temperature	-30	+25	+85	°C
T_J	Junction operating temperature	-30	+25	+125	°C

1. Minimum voltage does not include the required headroom for regulators.
2. V_{XX} is the supply voltage associated with the input or output pin to which the test voltage is applied.

3.3 DC power consumption

This section specifies DC power supply currents for the various IC operating modes (Table 3-3). Typical currents are based on IC operation at room temperature (+25°C) using default settings.

Table 3-3 DC power supply currents

Parameter		Condition	Min	Typ	Max	Units
I_ACTIVE	Supply current, active mode [1]	Post-PBS and SBL, unloaded regulators	–	4	8	mA
I_SLEEP	Supply current, sleep mode [2]	Sleep configuration, unloaded regulators	–	210	500	µA
IDD_OFF	Supply current, off mode [3]	Battery supply current, off mode	–	25	50	µA
IDD_CC	Supply current, off mode [4]	Coin cell supply current, off mode, XTAL off	–	10	25	µA
IDD_CC, rccal	Supply current, off mode [4]	Coin cell supply current, off mode, RCCAL	–	10	25	µA

1. I_ACTIVE is the total supply current from the main battery with the PMIC on, crystal oscillators on, temperature alarm active, internal infrastructure enabled, and the voltage regulators and signals enabled (FTSs in PWM, HFS in AMC, and LDOs in NPM).
2. I_SLEEP is the total supply current from the main battery with the PMIC on, temperature alarm active, internal infrastructure enabled, and the voltage regulators on with no load and the lowest power modes enabled. This specification only applies from -30°C to +60°C.
3. I_OFF is the total supply current from the main battery with the PMIC off. This only applies from -30°C to +60°C.
4. I_COIN is measured in PM6125 standalone configuration. It does not account for current consumed by external sources (such as PMI632 SYSOK PD on CBL_PWR_N).

3.4 Digital logic characteristics

All the PM6125/PM4250 digital I/O characteristics are specified in Table 3-4.

Table 3-4 Digital I/O characteristics

Parameter		Comments [1]	Min	Typ	Max	Units
V_{IH}	High-level input voltage	–	0.65 × V_{IO}	–	V_{IO} + 0.3	V
V_{IL}	Low-level input voltage	–	-0.3	–	0.35 × V_{IO}	V
V_{SHYS}	Schmitt hysteresis voltage	–	15	–	–	mV
I_L	Input leakage current [2]	V_{IO} = maximum, V_{IN} = 0 V to V_{IO}	-0.20	–	+0.20	µA
V_{OH}	High-level output voltage	I_{OUT} = I_{OH}	V_{IO} - 0.45	–	V_{IO}	V
V_{OL}	Low-level output voltage	I_{OUT} = I_{OL}	0	–	0.45	V
I_{OH}	High-level output current [3]	V_{OUT} = V_{OH}	3	–	–	mA
I_{OL}	Low-level output current [3]	V_{OUT} = V_{OL}	–	–	-3	mA
C_{IN}	Input capacitance [4]		–	–	5	pF

1. VIO is the supply voltage for the modem IC/PMIC interface (most PMIC digital I/Os).
2. GPIO pins comply with the input leakage specification only when configured as a digital input or set to the tri-state mode.
3. Output current specifications apply to all digital outputs, unless specified otherwise, and are superseded by specifications for specific pins (such as GPIO pins).
4. Input capacitance is guaranteed by design but is not 100% tested.

Table 3-5 GPIO_MV digital I/O characteristics

Parameter		LLimit	Typ	ULimit	Units	Test condition
Digital input	V_{IH}	0.65 × VDD	–	–	V	–
	V_{IL}	–	–	0.35 × VDD	V	–
	$V_{hysteresis}$	0.1 × VDD	–	–	mV	VDD ≤ 4 V
		0.4	–	–	V	VDD > 4 V
Digital output	V_{OH}	0.8 × VDD	–	VDD	V	I_{OH} = 0.3 mA
	V_{OL}	0	–	0.2 × VDD	mV	I_{OL}= 0.3 mA

Table 3-6 GPIO_MV load current drive

Supply voltage	V_{OL}, V_{OH}	Typical load current		
		Low strength driver	Medium strength driver	High strength driver
1.2 V	V_{OH} = 0.8 × VDD	0.15 mA	0.3 mA	0.6 mA
	V_{OL} = 0.2 × VDD			
1.8 V	V_{OH} = 0.8 × VDD	0.15 mA	0.3 mA	0.6 mA
	V_{OL} = 0.2 × VDD			
3.6 V	V_{OH} = 0.8 × VDD	0.25 mA	0.5 mA	1 mA
	V_{OL} = 0.2 × VDD			

GPIO may be able to drive higher than typical, but V_{OH} and V_{OL} levels are not guaranteed.

Table 3-7 GPIO_LV digital I/O characteristics

Parameter		LLimit	Typ	ULimit	Units	Test condition
Digital input	V_{IH} at VDD = 1.8 V	0.9	–	–	V	
	V_{IL} at VDD = 1.8 V	–	–	0.6	V	
	$V_{hysteresis}$ at VDD = 1.8 V	175	–	–	mV	
	V_{IH} at VDD = 1.2 V	0.78	–	–	V	
	V_{IL} at VDD = 1.2 V	–	–	0.39	V	
	$V_{hysteresis}$ at VDD = 1.2 V	120	–	–	mV	
Digital output	V_{OH}	0.8 × VDD	–	VDD	V	I_{OH} = 1.5 mA
	V_{OL}	0	–	0.2 × VDD	V	I_{OL} = 1.5 mA

Table 3-8 GPIO_LV load current drive

Supply voltage	V_{OL}, V_{OH}	Typical load current		
		Low strength driver	Medium strength driver	High strength driver
1.8 V	V_{OH} = 0.8 × VDD	1 mA	1.5 mA	2 mA
	V_{OL} = 0.2 × VDD			
1.2 V	V_{OH} = 0.8 × VDD	1 mA	1.5 mA	2 mA
	V_{OL} = 0.2 × VDD			

GPIO may be able to drive higher than typical, but V_{OH} and V_{OL} levels are not guaranteed.

3.5 Coin-cell charging

Coin-cell charging is enabled through software control and powered from VBAT. The on-chip charger is implemented using a programmable voltage source and a programmable series resistor. The modem IC reads the coin-cell voltage through the PMIC's analog multiplexer to monitor charging. Coin-cell charging performance is specified in Table 3-9.

Table 3-9 Coin-cell charging performance specifications

Parameter	Comments	Min	Typ	Max	Units
Target regulator voltage [1]	V_{IN} > 3.3 V, I_{CHG} = 100 µA	2.50	–	3.20	V
Target series resistance [2]		800	–	2100	Ω
Coin-cell charger voltage error	I_{CHG} = 0 µA	-5	–	+5	%
Coin-cell charger resistor error		-20	–	+20	%
Dropout voltage [3]	I_{CHG} = 2 mA	–	–	200	mV
Ground current, charger enabled	PMIC = off; VCOIN = open				
VBAT = 3.6 V, T = 27°C		–	4.5	–	µA
VBAT = 2.5 – 5.5 V		–	–	8	µA

1. Valid regulator voltage settings are 2.5, 3.0, 3.1, and 3.2 V.
2. Valid series resistor settings are 800, 1200, 1700, and 2100 Ω.
3. Set the input voltage (VBAT) to 3.5 V. Note the charger output voltage; call this value V_0. Decrease the input voltage until the regulated output voltage (V_1) drops 100 mV ($V_1 = V_0 - 0.1$ V). The voltage drop across the regulator under this condition is the dropout voltage ($V_{dropout}$ = VBAT - V_1).

3.6 Output power management

Output power management circuits include:

- Bandgap voltage reference circuit
- Fast-transient switched-mode power supply (FTS510) circuits
- High-frequency switched-mode power supply (HFS510) circuits
- Internal voltage-regulator connections
- LDO linear (LDO510) regulators

The PM6125/PM4250 device is the core PMIC, and is supplemented by the PMI632 interface device to provide all the regulated voltages needed for most wireless handset applications. Independent regulated power sources are required for various electronic functions to avoid signal corruption between diverse circuits, to support power-management sequencing, and to meet different voltage-level requirements.

A total of 32 programmable voltage regulators are provided by the PM6125/PM4250 device, with all outputs derived from a common bandgap reference circuit. Each regulator can be set to a low-power mode for power savings.

A high-level summary of all regulators is listed in Table 3-10 and Table 3-11.

Table 3-10 PM6125 SMPS regulator summary

Regulator	Circuit type	I_{RATED} (mA)	Default on booting voltage (V)	Specified range (V)	Programmable range (V)	Default on	Expected use (SM6125)
S1	FTS510 SMPS	4000	0.8	0.4–1.14	0.32–1.352	Y	APC0, APC1
S2	FTS510 SMPS	4000	0.8	0.4–1.14	0.32–1.352	Y	APC0, APC1
S3	FTS510 SMPS	4000	0.8	0.4–1.076	0.32–1.352	Y	CX + MSS
S4	FTS510 SMPS	4000	0.8	0.4–1.076	0.32–1.352	Y	CX + MSS
S5	HFS510 SMPS	4000	0.856	0.57–0.992	0.32–2.04	Y	MX, EBI_VDDIO, WCSS_MX, LPDDR4x_VDDQ (Subreg), WCSS_CX (Subreg)
S6	HFS510 SMPS	2000	1.352	0.936–1.422	0.32–2.04	Y	MV sub regulation
S7	HFS510 SMPS	2500	2.04	1.972–2.040	0.32–2.04	Y	HV sub regulation
S8	FTS510 SMPS	3000	1.128	1.096–1.304	0.32–1.352	Y	LV sub regulation, LPDDR4x_VDD2, SDR VDDPX_1

Table 3-11 PM6125 linear voltage regulator summary

Regulator	Circuit type	I_{RATED} (mA)	Default on booting voltage (V) [1]	Specified range (V)	Programmable range (V)	Default on	Expected use (SM6125)
L1	N600	600	1.2	1.178–1.25	0.312–1.304	N	QLN (1.2), QDL (1.2), SDR (1.2)
L2	N300	300	1.0	0.95–1.05	0.312–1.304	N	SDR (A1V)
L3	N600	600	1.0	0.97–1.06	0.312–1.304	Y	SDR (D1V)
L4	N300	300	0.928	0.88–0.97	0.312–1.304	Y	EBIPLL (0.93 V), USB2 (A0P9), USB3 (A0P9), QLINK, QREF, MIPI_DSI, UFS (0.9)
L5	MV P50	50	2.96	1.65–3.1	1.504–3.544	Y	PX2
L6	N1200	1200	0.624	0.57–0.65	0.312–1.304	Y	LPDDR4x_VDDQ (0.6), VDDIO_EBI, VDDIO_CK_EBI
L7	N600	600	0.928	0.875–0.975	0.312–1.304	Y	USB 0.9 V
L8	N300	300	0.664	0.4–0.728	0.312–1.304	Y	VDD_WCSS_CX (0.73)
L9 [2]	LV P600	600	1.8	1.8	1.504–2.000	Y	LPDDR4x_VDD1 (1.8), VDD_PX3, VDD_PX7, WCD (1.8 V), WCN (1.8 V), Sensors (1.8 V), RFFE (1.8 V), WSA (1.8 V)

Table 3-11 PM6125 linear voltage regulator summary (cont.)

Regulator	Circuit type	I_{RATED} (mA)	Default on booting voltage (V) [1]	Specified range (V)	Programmable range (V)	Default on	Expected use (SM6125)
L10	LV P150	150	1.8	1.7–1.89	1.504–2.000	Y	USB2 (1.8 V), USB3.1 (1.8 V), QFPROM (1.8 V), UFS (1.8 V), VDDPX_11
L11	LV P600	600	1.8	1.7–1.95	1.504–2.000	Y	EMMC1_VDDQ (1.8 V), UFS (1.8 V)
L12	LV P300	300	1.8	1.7–1.9	1.504–2.000	Y	Camera_sensors (1.8 V), USB_Redriver, MIPI_VDDIO (1.8 V)
L13	LV P300	300	1.8	1.721–1.829	1.504–2.000	N	SDR (1.8 V), QPM (1.8 V), QDM (1.8 V)
L14	LV P600	600	1.8	1.7–1.9	1.504–2.000	N	WCD_VDD_BUCK (1.8 V)
L15	MV P150	150	3.128	2.93–3.23	1.504–3.544	Y	USB3 (3.1)
L16	LV P150	150	1.8	1.7–1.9	1.504–2.000	N	WCN (1.8 V), WCSS_PLL
L17	N300	300	1.304	1.245–1.3	0.312–1.304	N	WCN (1.3 V), WCSS_ADC_DAC (1.3 V)
L18	N300	300	1.232	1.19–1.26	0.312–1.304	Y	EBI_PLL (1.23), MIPI_CSI, MIPI_DSI (1.23), VDDPX_10
L19	MV P150	150	1.8	1.65–2.95	1.504–3.544	N	UIM1 (1.8), PX5
L20	MV P150	150	1.8	1.65–2.95	1.504–3.544	N	UIM2 (1.8), PX6
L21	MV P600	600	2.704	2.6–2.85	1.504–3.544	N	ASDIV_SW, QAT (2.7), RF_Switch
L22 [3]	MV P600	600	2.96	2.95–3.3	1.504–3.544	Y	MEM_SD_MMC_VDD (2.96)
L23	MV P600	600	3.304	3.2–3.4	1.504–3.544	N	WCN (3.3)
L24	MV P600	600	2.96	2.7–3.6	1.504–3.544	Y	EMMC1_VCC (2.96)/UFS_MEM (2.96)

1. All regulators have default voltage settings, whether they default on or not; the voltage and state depends upon the PBS configuration.
2. L9 powers internal circuits that are limited to 1.8V operation; its programmed voltage should not be changed, and it should not be turned off.
3. LDO L22 would be able to provide current of 800 mA to support SDR104 mode. The regulation specification of 2% would not be met. Since minimum voltage needed at SD card is only 2.7 V, accuracy of 8.4% is sufficient. The LDO L22 can provide current of 600 mA meeting all regulation specification.

Table 3-12 PM4250 SMPS regulator summary

Regulator	Circuit type	I_{RATED} (mA)	Default on booting voltage (V)	Specified range (V)	Programmable range (V)	Default on	Expected use (SM4250/SM6115)
S1	FTS510 SMPS	4000	0.8	0.46–1.15	0.32–1.352	Y	APC0, APC1
S2	FTS510 SMPS	4000	0.8	0.46–1.15	0.32–1.352	Y	APC0, APC1
S3	FTS510 SMPS	4000	0.8	0.4–1.08	0.32–1.352	Y	CX + MSS
S4	FTS510 SMPS	4000	0.8	0.4–1.08	0.32–1.352	Y	CX + MSS
S5	HFS510 SMPS	4000	0.848	0.45–0.99	0.32–2.04	Y	MX, EBI_VDDIO, WCSS_MX, LPDDR4x_VDDQ (Subreg), WCSS_CX (Subreg)
S6	HFS510 SMPS	2000	1.352	0.3–1.45	0.32–2.04	Y	MV sub regulation
S7	HFS510 SMPS	2500	2.04	1.28–2.08	0.32–2.04	Y	HV sub regulation
S8	FTS510 SMPS	3000	1.128	1.06–1.3	0.32–1.352	Y	LV sub regulation, LPDDR4x_VDD2, VDDPX_1

Table 3-13 PM4250 linear voltage regulator summary

Regulator	Circuit type	I_{RATED} (mA)	Default on booting voltage (V) [1]	Specified range (V)	Programmable range (V)	Default on	Expected use (SM4250/SM6115)
L1	N600	600	1.0	0.95–1.15	0.312–1.304	Y	WTR 1 V
L2	N300	300	0.848	0.488–0.988	0.312–1.304	Y	LPI_MX
L3	N600	600	0.8	0.4–1.08	0.312–1.304	Y	LPI_CX
L4	N300	300	0.928	0.488–1.0	0.312–1.304	Y	EBIPLL (0.93 V), USB2 (A0P9), USB3 (A0P9), MIPI_DSI, UFS (0.9), USB 0.9 V
L5	MV P50	50	2.96	1.65–3.05	1.504–3.544	Y	PX2
L6	N1200	1200	0.624	0.57–0.65	0.312–1.304	Y	LPDDR4x_VDDQ(0.6), VDDIO_EBI, VDDIO_CK_EBI
L7	N600	600	1.256	1.2–1.3	0.312–1.304	Y	UIM/SDC VBIAS
L8	N300	300	0.664	0.4–0.728	0.312–1.304	Y	VDD_WCSS_CX (0.73)

Table 3-13 PM4250 linear voltage regulator summary (cont.)

Regulator	Circuit type	I_{RATED} (mA)	Default on booting voltage (V) [1]	Specified range (V)	Programmable range (V)	Default on	Expected use (SM4250/SM6115)
L9 [2]	LV P600	600	1.8	1.8–2.0	1.504–2.000	Y	LPDDR4x_VDD1 (1.8), VDD_PX3, VDD_PX7, WCD (1.8 V), WCN (1.8 V), Sensors (1.8 V), RFFE (1.8 V), WSA (1.8 V)
L10	LV P150	150	1.8	1.7–1.9	1.504–2.000	Y	BBCLK_VREG
L11	LV P600	600	1.8	1.7–1.95	1.504–2.000	Y	EMMC1_VDDQ(1.8),UFS(1.8)
L12	LV P300	300	1.8	1.62–1.98	1.504–2.000	Y	WCN (1.8 V), WCSS_PLL, MIPI_VDDIO (1.8)
L13	LV P300	300	1.8	1.5–1.95	1.504–2.000	Y	WTR 1.8 V
L14	LV P600	600	1.8	1.7–1.9	1.504–2.000	N	WCD_VDD_BUCK (1.8)
L15	MV P150	150	3.128	2.92–3.23	1.504–3.544	Y	USB3 (3.1)
L16	LV P150	150	1.8	1.7–1.9	1.504–2.000	N	WCSS PLL, WCN XO
L17	N300	300	1.304	1.15–1.39	0.312–1.304	N	WCN (1.3), WCSS_ADC_DAC (1.3)
L18	N300	300	1.232	1.1–1.312	0.312–1.304	Y	EBI_PLL (1.23), MIPI_CSI, MIPI_DSI (1.23), VDDPX_10
L19	MV P150	150	1.8	1.62–3.3	1.504–3.544	N	UIM1 (1.8), PX5
L20	MV P150	150	1.8	1.62–3.3	1.504–3.544	N	UIM2 (1.8), PX6
L21	MV P600	600	2.704	2.4–3.6	1.504–3.544	N	ASDIV_SW, QAT(2.7), RF_Switch
L22 [3]	MV P600	600	2.96	2.95–3.3	1.504–3.544	Y	MEM_SD_MMC_VDD (2.96)
L23	MV P600	600	3.304	3.2–3.4	1.504–3.544	N	WCN (3.3)
L24	MV P600	600	2.96	2.7–3.6	1.504–3.544	Y	EMMC1_VCC (2.96)/UFS_MEM (2.96)

1. All regulators have default voltage settings, whether they default on or not; the voltage and state depends upon the PBS configuration.
2. L9 powers internal circuits that are limited to 1.8V operation; its programmed voltage should not be changed, and it should not be turned off.
3. LDO L22 would be able to provide current of 800 mA to support SDR104 mode. The regulation specification of 2% would not be met. Since minimum voltage needed at SD card is only 2.7 V, accuracy of 8.4% is sufficient. The LDO L22 can provide current of 600 mA meeting all regulation specification.

3.6.1 Reference circuit

All PMIC regulator circuits, and some other internal circuits, are driven by a common, on-chip voltage reference circuit. An on-chip series resistor supplements an off-chip 0.1 µF bypass capacitor at the REF_BYP pin to create a low-pass function that filters the reference voltage distributed throughout the device.

NOTE: Do not load the REF_BYP pin. Use a GPIO configured as an analog output if the reference voltage is needed off-chip.

Applicable voltage reference performance specifications are given in Table 3-14.

Table 3-14 Voltage reference performance specifications

Parameter	Comments	Min	Typ	Max	Units
Nominal internal VREF	At REF_BYP pin	–	1.250	–	V
Output voltage deviations					
Normal operation	Over temperature only, -20°C to +70°C	-0.20	–	0.20	%
Normal operation	All operating conditions	-0.35	–	0.35	%
Sleep mode	All operating conditions	-1.00	–	1.00	%

3.6.2 FT-SMPS

The PM6125/PM4250 device includes five FTS510 SMPS circuits. All FT-SMPS circuits can be combined for multi-phase operation. PWM and pulse-skipping modes are supported. It also supports a retention mode, which is an ultra-low-power state allowing significant efficient improvements in sleep mode. Pertinent target performance specifications are given in Table 3-15.

Table 3-15 FT-SMPS performance specifications

Parameter	Comments	Min	Typ	Max	Units
General characteristics					
Operating input voltage range	Specification-compliant V_{IN} window	2.5	–	4.8	V
Extended input voltage range	Short duration VIN excursions beyond normal operating voltage range (< 1 ms)	2.0	–	5.25	V
Output voltage range		0.300	–	1.352	V
NPM, any number of phases					
Rated steady-state load current per phase	Full-sized power stage; alternate-sized PS ratings to be added once defined	4.0	–	–	A
DC output voltage accuracy	VREG ≥ 0.8 VREG < 0.8 -30°C to 125°C	-2 -16	– –	2 16	% mV
DC output voltage accuracy at trimmed set point	VREG = trimmed set point -30°C to 125°C	-1.8 -14.4	– –	1.8 25	% mV

Table 3-15 FT-SMPS performance specifications (cont.)

Parameter	Comments	Min	Typ	Max	Units
Ripple voltage, with 3 × 10 µF + 1 × 4.7 µF	Unloaded through I_{RATED}, 20 MHz bandwidth limited	–	15	50	mVpp
Line transient response	GSM burst induced line transient represented by: Rbat = 350 mΩ, Istep = 2 A with 10 µs slew VPH_PWR capacitance = 100 µF	–	–	20	mVpp
Ground current					
Ground current NPM	No load, single phase No load, multiphase	– –	0.150 0.170	0.200 0.220	mA mA
Ground current Retention mode (commanded)	No load, any number of phases	–	5	–	µA
NPM load transient, any number of phases, unless otherwise stated					
Load transient dip/bump voltage disturbance: full performance (2 A load step) configuration with 3 × 10 µF + 1 × 4.7 µF per phase	Any phase count 2 A load step per phase, transient slew ~100 ns 1 V output test condition	-50	–	80	mV
Load transient regulation window: 0.5 A load step for VDDQ V_{OUT} AC compliance	LPDDR4x/y VDDQ: single phase, 0.5 A load step, transient slew ~100 ns	0.575	0.6	0.645	V
Retention mode, any number of phases					
DC output accuracy, retention mode	DC error relative to programmed VSET -30°C to 125°C	-40	–	70	mV
Rated load current, retention mode		–	10	–	mA
Rated transient step, retention mode		–	5	–	mA
Ripple voltage, retention mode	No load to the maximum rated RM loading (DC). 20 MHz bandwidth limited	–	40	70	mVpp
Other general characteristics					
Enable warmup time	Time from enable to start of V_{OUT} ramping	–	20	–	µs
Enable overshoot	4.8 mV/µs setting. Soft start stepper rate must be programmed to draw less than 75% of I_{RATED}.	–	–	8	mV
Soft start and voltage stepper slew rate	Actual voltage ramping slew rate (not counting the warmup time)	4.8	19.2	38.4	mV/µs
Peak output impedance	1 kHz to 1 MHz	–	–	40	mΩ
Discharge impedance, strong pull-down		–	150	–	Ω

Table 3-15 FT-SMPS performance specifications (cont.)

Parameter	Comments	Min	Typ	Max	Units
Discharge impedance, weak pull-down		–	10	–	kΩ
Discharge impedance, leak pull-down		–	500	–	kΩ

The following efficiency plots are considered typical plots for reference with nominal settings and operating conditions.

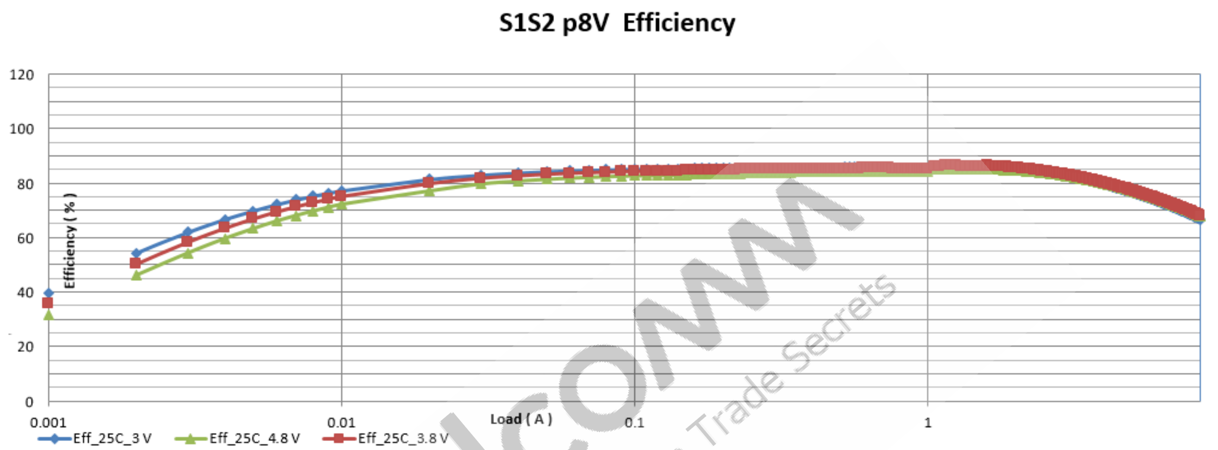


Figure 3-1 S1-S2 efficiency, auto mode

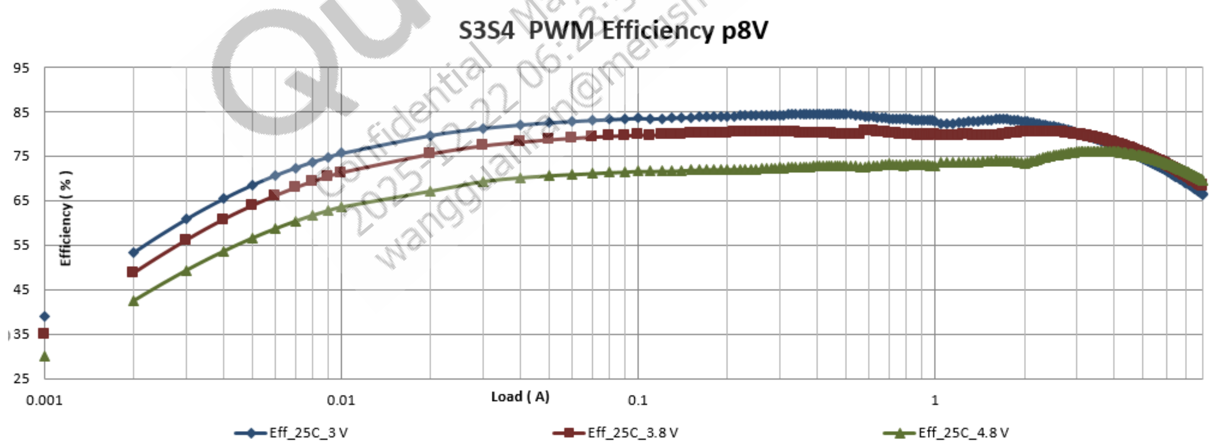


Figure 3-2 S3-S4 efficiency, force PWM mode

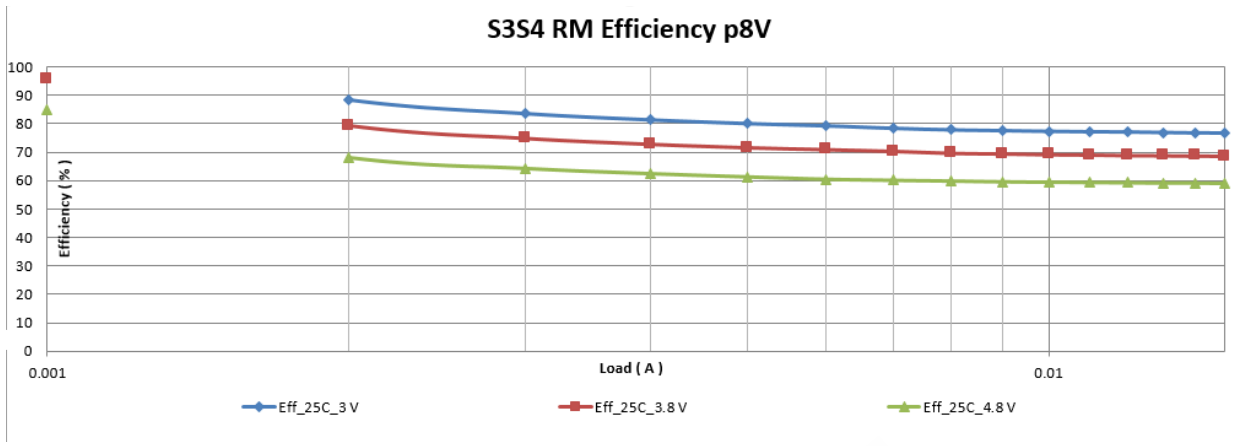


Figure 3-3 S3-S4 efficiency, retention mode

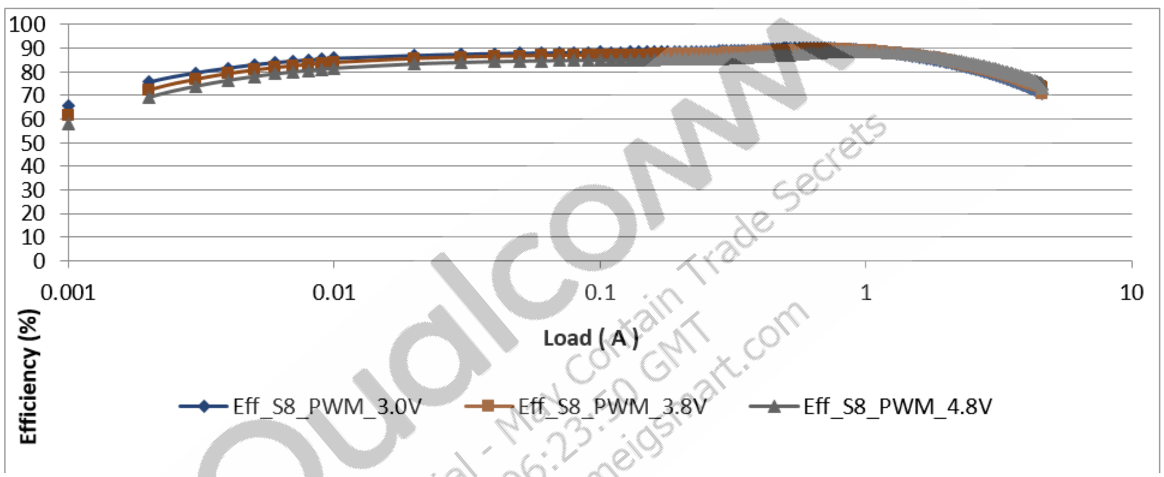


Figure 3-4 S8 efficiency, force PWM mode

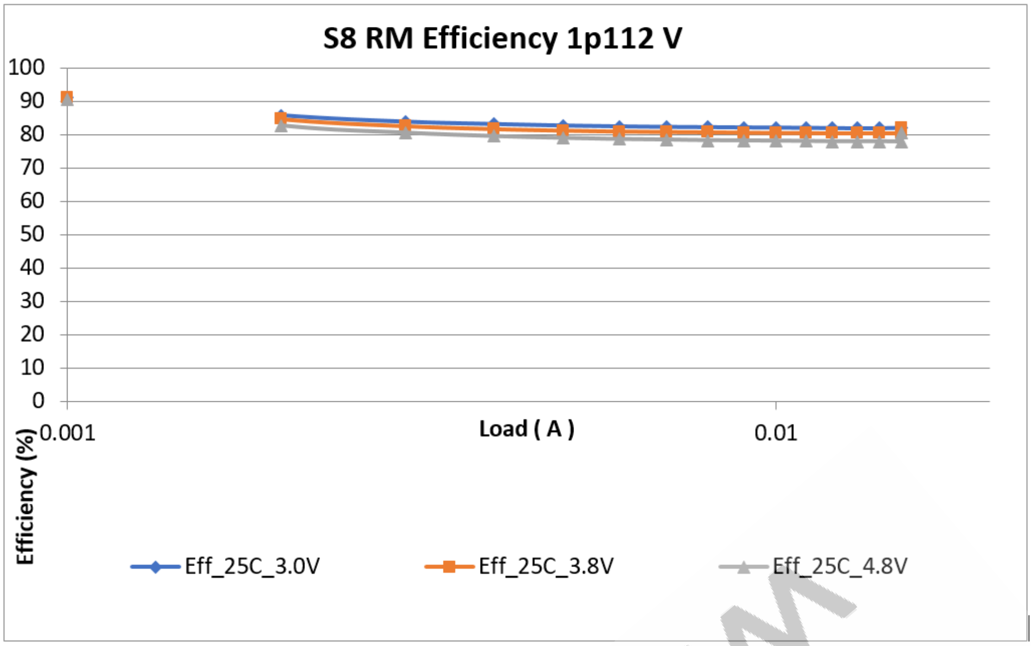


Figure 3-5 S8 efficiency, retention mode

3.6.3 HF-SMPS

The PM6125/PM4250 device includes three HFS510 SMPS circuits. It supports PWM and PFM modes, and the automatic transition between PWM and PFM modes, depending on the load current. It also supports a retention mode, which is an ultra-low power state allowing significant efficient improvements in sleep. Pertinent performance specifications are given in Table 3-16.

Table 3-16 HFS510 buck specifications

Parameter	Comments	Min	Typ	Max	Units
Low/high-power mode					
Output voltage operating range		0.32	–	2.04	V
Enable overshoot (voltage upon buck enabling)		–	–	70	mV
Enable settling time (turning on an off regulator)	From enable to within 1% of the final value, no load	–	–	500	µs
Ground current, no load	PWM mode	–	550	–	µA
	PFM mode	–	50	–	µA
	Retention mode	–	2	–	µA
Power supply ripple rejection ratio (PSRR)	50 Hz	–	40	–	dB
	1 kHz	–	40	–	dB
	100 kHz	–	28	–	dB
	1 MHz	–	28	–	dB

Table 3-16 HFS510 buck specifications (cont.)

Parameter	Comments	Min	Typ	Max	Units
Discharge time to 10% of the output voltage	Nominal load cap 10 μF (effective), initial voltage = 2.04 V. Strong pull-down enabled.	–	0.5	3	ms
PWM mode					
DC accuracy (over voltage, normal mode, over process, load, temperature, and line regulation)	$V_{OUT} \geq 0.8$ V $0.32 \leq V_{OUT} < 0.8$ V	-2 -16	0 0	2 16	% mV
Load transient response [1]	PWM mode, 200 mA load attack or release	-50	–	70	mV
Ripple voltage in PWM, continuous switching	Tested using 20 MHz bandwidth limit	–	10	20	mVpp
Ripple voltage in PWM, light-load pulse skipping	Tested using 20 MHz bandwidth limit at 40 mA load	–	20	50	mVpp
PFM and retention modes					
Rated load current	PFM and retention mode	200	–	--	mA
DC accuracy PFM mode (over voltage, over process, load, temperature, and line regulation)	$V_{OUT} \geq 0.8$ V, $I_{RATED}/2$ $0.32 \leq V_{OUT} < 0.8$ V, 100 mA	-2 -16	– –	4 40	% mV
Ripple voltage in PFM mode	PFM low-current mode, measured using 20 MHz bandwidth	–	–	70	mVpp
Retention mode voltage window		-40	–	70	mV
Mode transition voltage transient PWM ←→ PFM		-50	–	+70	mV
Retention mode load transient		-50	–	+80	mV
Auto mode					
Load transient response [1]	400 mA load step	-50	–	+80	mV

1. Transient response optimizations are configured in SBL. The booting configuring prior to SBL optimization is designed to support initial system loading.

The following efficiency plots are considered typical plots for reference with nominal settings and operating conditions.

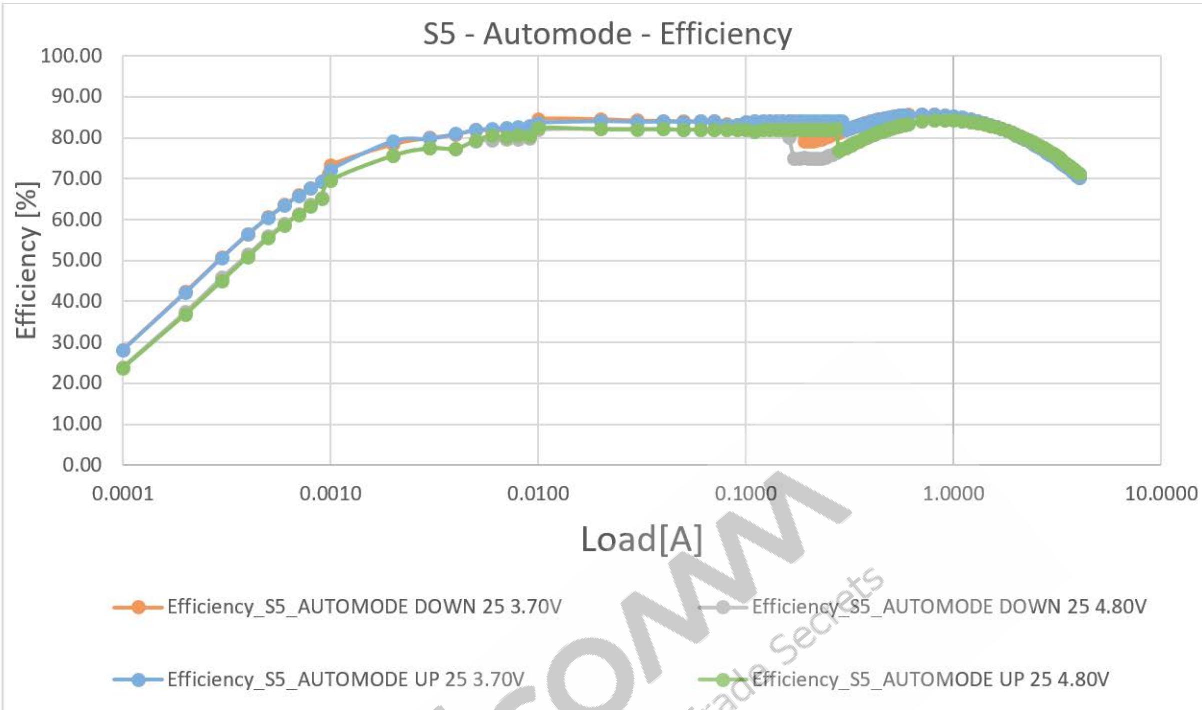


Figure 3-6 S5 efficiency auto mode

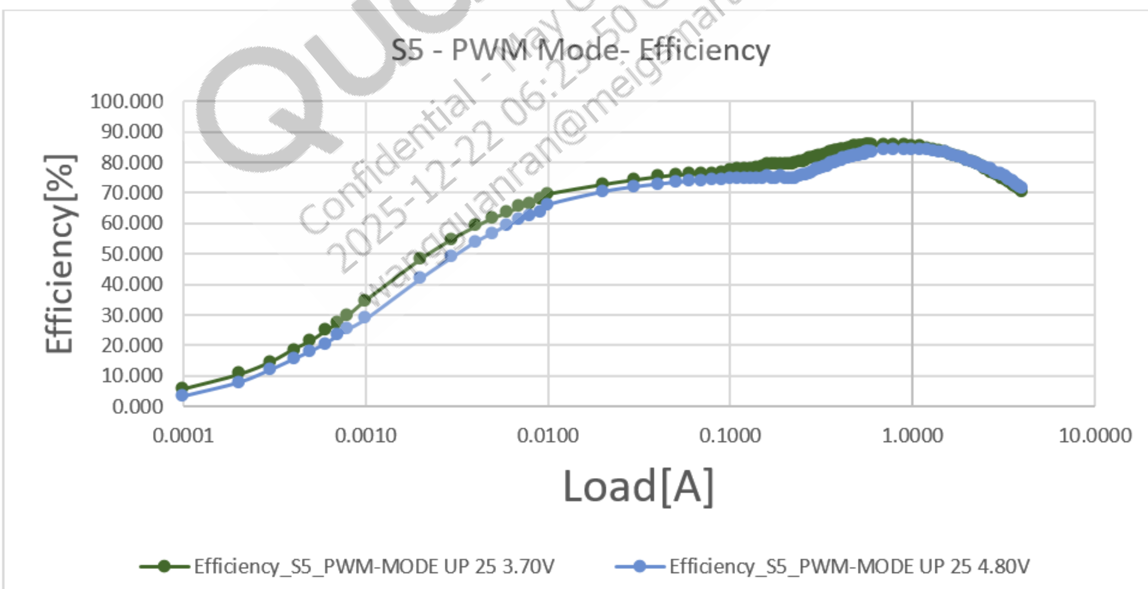


Figure 3-7 S5 efficiency, force PWM mode

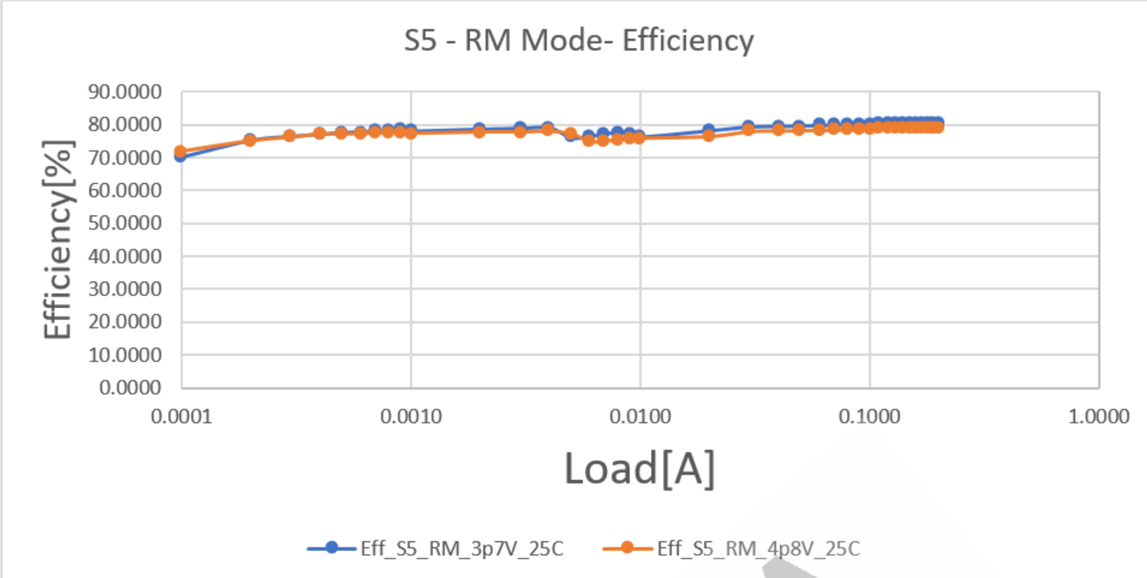


Figure 3-8 S5 efficiency, retention mode

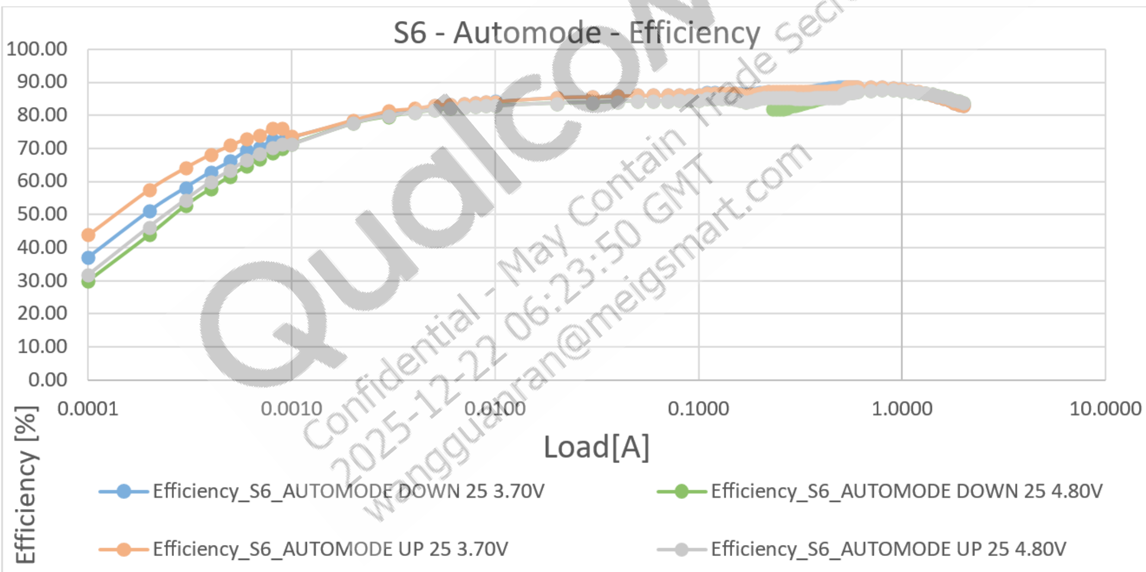


Figure 3-9 S6 efficiency, auto mode

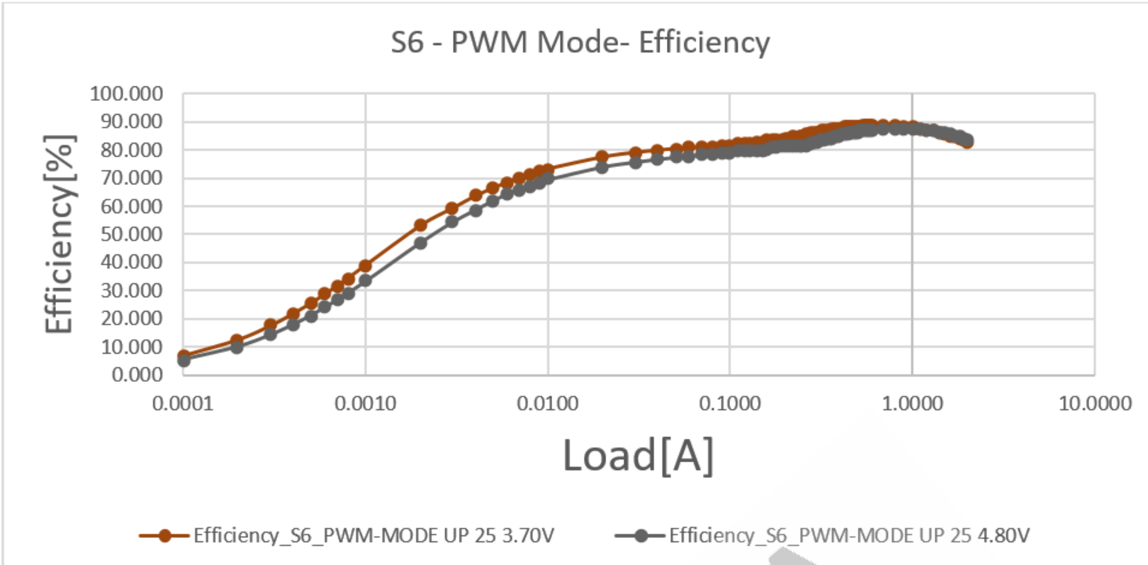


Figure 3-10 S6 efficiency, force PWM mode

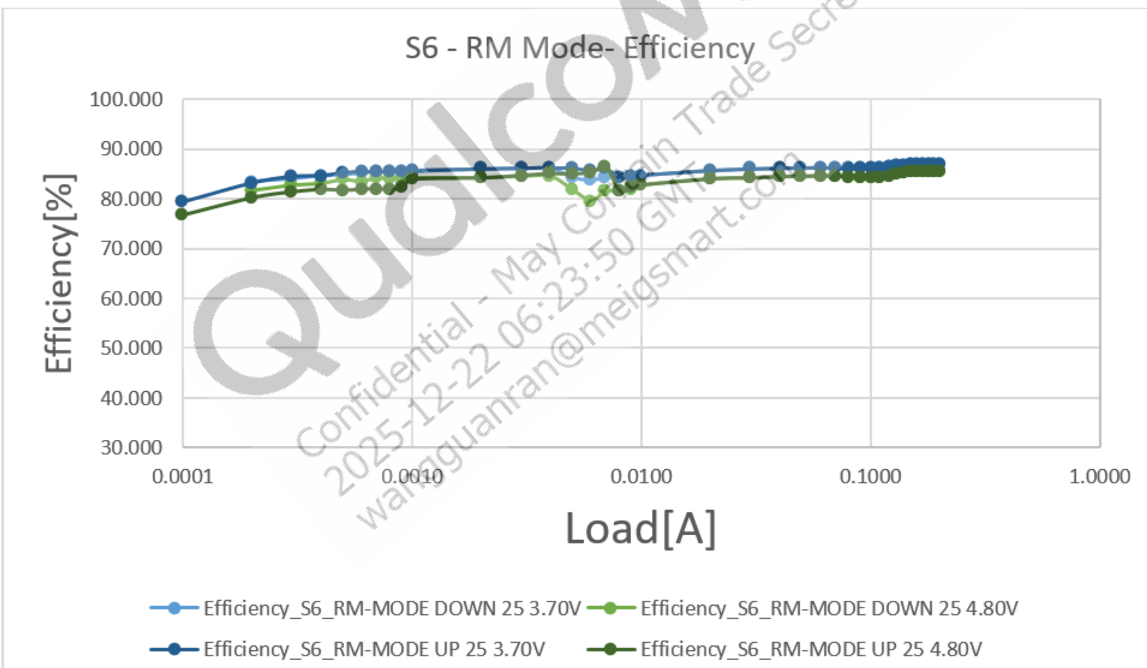


Figure 3-11 S6 efficiency, retention mode

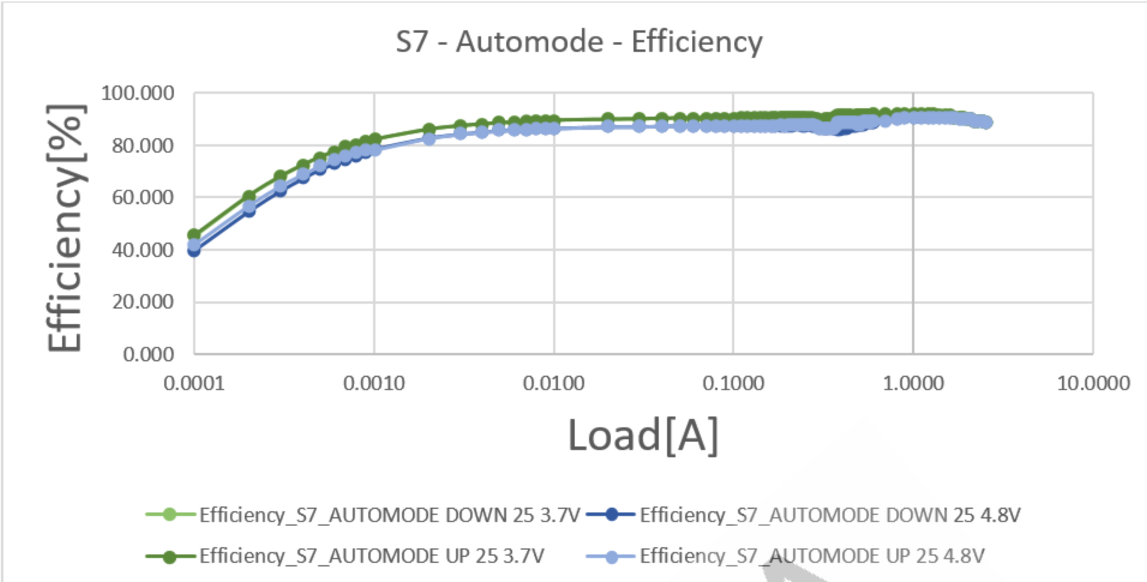


Figure 3-12 S7 efficiency, auto mode

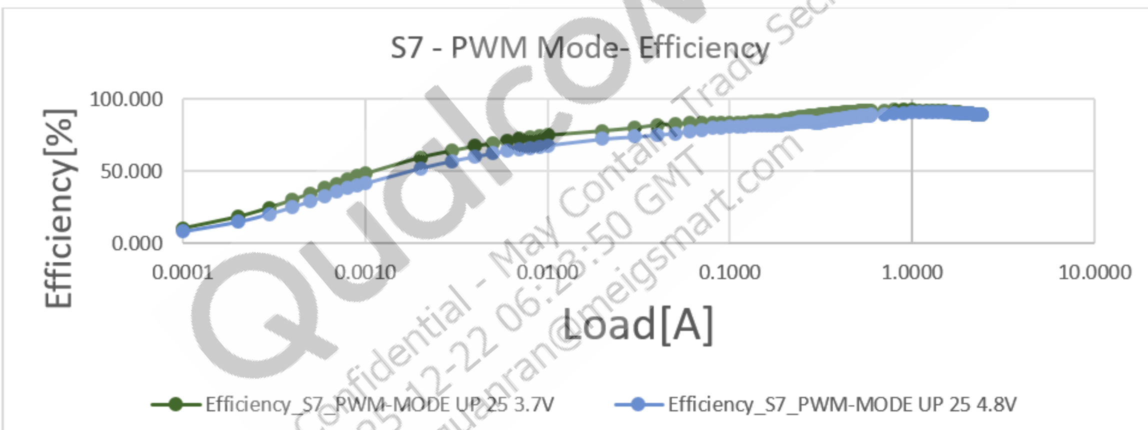


Figure 3-13 S7 efficiency, force PWM mode

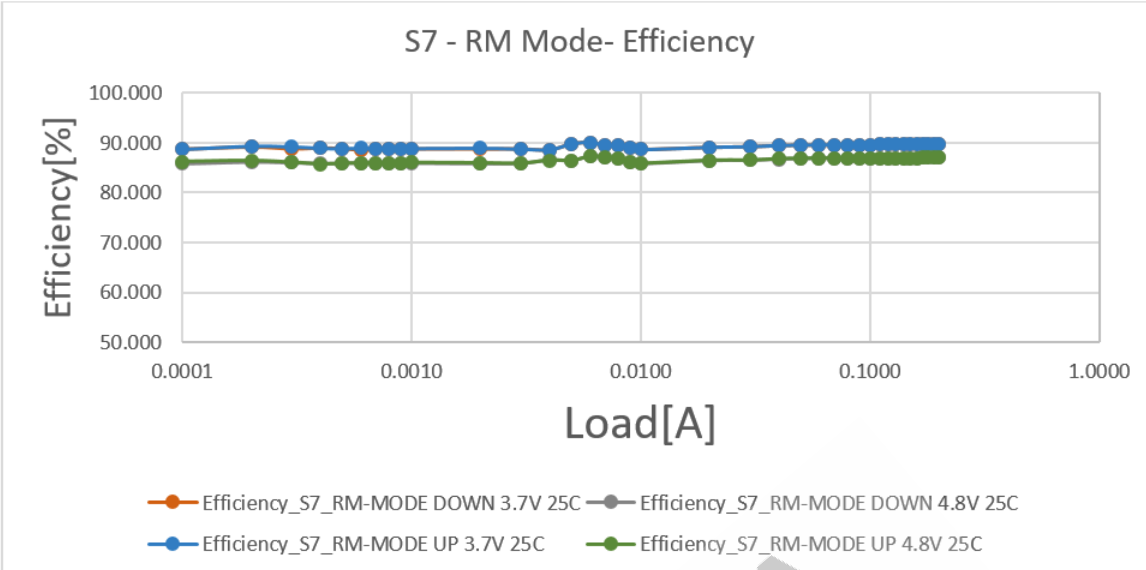


Figure 3-14 S7 efficiency, retention mode

3.6.4 Linear regulators

24 low dropout (LDO) linear regulator designs are implemented within the PMIC.

NOTE The specifications in Table 3-17 for LV PMOS, MV PMOS, and NMOS LDOS require that the external component and PCB routing parasitics be in compliance with the targeted values listed in the *Understanding Low-Dropout (LDO) Regulators Application Note* (80-VT310-125).

Table 3-17 LDO regulator specifications

Parameter	Comments [1]		Min	Typ	Max	Units	Note
Input voltage range	NMOS LV PMOS (V_{IN} > 2.1 V degrades device reliability) MV PMOS		0.32 1.8 2.5	– – –	1.4 2.04 5.5	V V V	
Output voltage range	NMOS (UL = 1.272 V in retention) LV PMOS MV PMOS		0.32 1.504 1.504	– – –	1.304 2.0 3.544	V V V	
V_{OUT} step size			–	8	–	mV	
Normal power mode							
Rated load current	Maximum load current at which all specifications are met below headroom. NMOS and LV PMOS: HR = 100 mV (PS1) MV PMOS: HR = 500 mV Higher load current is possible at increased headroom.	LV and MV P50 LV and MV P150 LV and MV P300, N300 LV and MV P600, N600 N1200	50 150 300 600 1200	– – – – –	– – – – –	mA mA mA mA mA	2
Overall DC error at default voltage (includes load and line regulation, temperature, VBAT, MBG variation, and trim error)	Measured at Cout. $I_{RATIO} = I_{LOAD}/I_{RATED}$ NMOS & LV PMOS: HR = 100 mV (PS1) and HR = 160 mV (PS2). MV PMOS: HR = 500 mV Format: X + Y + Z X = Performance at LDO sense point Y = On-die routing DCR contribution Z = PCB routing DCR contribution	NMOS ($V_{OUT} \geq 0.8$ V)	- 1.3% - (12 mV + 30 mV) × I_{RATIO}	–	+1.3%	$\%V_{OUT}$	
		LV PMOS	- 1.2% - (6 mV + 15 mV) × I_{RATIO}	–	+1.2%	$\%V_{OUT}$	
		MV PMOS	- 1% - (6 mV + 15 mV) × I_{RATIO}	–	+1%	$\%V_{OUT}$	

Table 3-17 LDO regulator specifications (cont.)

Parameter	Comments [1]		Min	Typ	Max	Units	Note
Overall DC error at non-default voltage (Includes load and line regulation, temperature, VBAT, MBG variation and trim error)	Measured at Cout. $I_{RATIO} = I_{LOAD}/I_{RATED}$ NMOS & LV PMOS: HR = 100 mV (PS1) and HR = 160 mV (PS2). MV PMOS: HR = 500 mV Format: X + Y + Z X = Performance at LDO sense point Y = On-die routing DCR contribution Z = PCB routing DCR contribution	NMOS (For $V_{OUT} \geq 0.5$ V)	- 2% - (12 mV + 30 mV) × I_{RATIO}	–	+2%	$\%V_{OUT}$	
		NMOS (For $V_{OUT} < 0.5$ V, upto $I_{RATED}/4$)	- (10 + 3 + 8)	–	+10	mV	
		LV PMOS	- 2% - (6 mV + 15 mV) × I_{RATIO}		+2%	$\%V_{OUT}$	
Load transient undershoot, overshoot	Max load step of 0.5 × I_{RATED} in 1 µs, with baseline current of 0.01 × I_{RATED} or higher. Compared to final settled value. NMOS and LV PMOS: HR = 100mV (PS1) MV PMOS: HR = 500 mV	NMOS	–40		+70	mV	
		LV PMOS	-40		+70	mV	
		MV PMOS	-50		+70	mV	
Startup settling time	To within 1% of final value. UL is with maximum C_{OUT}. NMOS startup time depends on V_{SET} and stepper rate setting.	NMOS LV PMOS (V_{OUT} = 1.8 V) MV PMOS (V_{OUT} = 3.3 V)	– – –	140 200 –	300 300 550	µs	3
Startup in-rush current	During start-up. Variation due to bulk capacitor size.	N300	–	40	400	mA	
		N600	–	80	800	mA	
		N1200	–	160	900	mA	
		LV P150	–	50	250	mA	
		LV P300	–	50	400	mA	
		LV P600	–	170	700	mA	
		MV P50	–	30	250	mA	
		MV P150	–	25	300	mA	
		MV P300	–	40	400	mA	
		MV P600	–	150	800	mA	

Table 3-17 LDO regulator specifications (cont.)

Parameter	Comments [1]		Min	Typ	Max	Units	Note
Dropout voltage	NMOS. From parent buck C_{OUT} to LDO sense point. Load at I_{RATED}	PS1 – for Vph ≥ 3 V	–	–	54	mV	4
		PS1 – for Vph < 3 V	–	–	65	mV	
		PS2 – for Vph ≥ 3 V	–	–	85	mV	
		PS2 – for Vph < 3 V	–	–	95	mV	
	LV PMOS	PS1	–	–	54	mV	
		PS2	–	–	85	mV	
	MV PMOS. From parent buck C_{OUT} to LDO sense point. Load at I_{RATED}		–	–	300	mV	
Load regulation	$I_{LOAD} = I_{RATED}/100$ to I_{RATED}. Measured at Cout. $I_{RATIO} = I_{LOAD}/I_{RATED}$ NMOS and LV PMOS: HR = 100 mV (PS1) and HR = 160 mV (PS2). MV PMOS: HR = 500 mV Format: X+Y+Z X = Performance at LDO sense point Y = On-die routing DCR contribution Z = PCB routing DCR contribution	NMOS	–	–	0.3% + (12 mV +30 mV) × I_{RATIO}		
		LV PMOS	–	–	0.3% + (6 mV+15 mV) × I_{RATIO}		
		MV PMOS	–	–	0.3% + (6 mV+ 15 mV) × I_{RATIO}		
Line regulation	Based on VBAT from 3 V to 5 V. For $V_{out} \geq 0.5$ V NMOS and LV PMOS: HR = 100 mV (PS1) and HR = 160 mV (PS2). MV PMOS: HR = 500 mV		–	–	0.1	%/V	
Over current protection detection threshold (I_{OCP})	Threshold for OCP detection. OCP interrupt and self-shutdown are based on tripping this threshold.	N300	400	550	900	mA	
		N600	800	1100	1800	mA	
		N1200	1400	2400	3300	mA	
		LV P150	225	375	450	mA	
		LV P300	500	800	1200	mA	
		LV P600	1100	1500	2200	mA	
		MV P50	75	100	200	mA	
		MV P150	225	325	525	mA	
		MV P300	450	650	950	mA	
		MV P600	900	1200	2200	mA	

Table 3-17 LDO regulator specifications (cont.)

Parameter	Comments [1]		Min	Typ	Max	Units	Note
Output noise density	Measured at C_{OUT}. LDO contribution only, assumes clean V_{IN}.	NMOS: HR = 100 mV (PS1)					
		100 Hz ≤ f ≤ 1 kHz	–	3	–	$\mu V/\sqrt{Hz}$	
		1 kHz ≤ f ≤ 10 kHz	–	1.5	–		
		10 kHz ≤ f ≤ 100 kHz	–	1.2	–		
		100 kHz ≤ f ≤ 1 MHz	–	0.5	–		
		LV PMOS: HR = 100 mV (PS1)					
		100 Hz ≤ f ≤ 1 kHz	–	5	–	$\mu V/\sqrt{Hz}$	
		1 kHz ≤ f ≤ 10 kHz	–	1.7	–		
		10 kHz ≤ f ≤ 100 kHz	–	0.65	–		
		100 kHz ≤ f ≤ 1 MHz	–	0.25	–		
		MV PMOS: HR = 500 mV					
		100 Hz ≤ f ≤ 1 kHz	–	11	–	$\mu V/\sqrt{Hz}$	
		1 kHz ≤ f ≤ 10 kHz	–	3.5	–		
		10 kHz ≤ f ≤ 100 kHz	–	1.2	–		
		100 kHz ≤ f ≤ 1 MHz	–	0.7	–		
PSRR	$I_{LOAD} = I_{RATED}$. From V_{IN} to V_{OUT}. Measured at C_{OUT}.	NMOS: HR = 100 mV (PS1)					
		50 Hz to 1 kHz	–	55	–	dB	
		1 kHz to 10 kHz	–	40	–	dB	
		10 kHz to 100 kHz	–	30	–	dB	
		100 kHz to 1 MHz	–	20	–	dB	
		LV PMOS: HR = 100 mV (PS1)					
		50 Hz to 1 kHz	–	50	–	dB	
		1 kHz to 10 kHz	–	30	–	dB	
		10 kHz to 100 kHz	–	25	–	dB	
		100 kHz to 1 MHz	–	15	–	dB	
		MV PMOS: HR = 500 mV					
		50 Hz to 1 kHz	–	45	–	dB	
		1 kHz to 10 kHz	–	35	–	dB	
		10 kHz to 100 kHz	–	30	–	dB	
		100 kHz to 1 MHz	–	20	–	dB	

Table 3-17 LDO regulator specifications (cont.)

Parameter	Comments [1]		Min	Typ	Max	Units	Note
Ground current, no load	Measured at the battery. I_Q may be much higher if LDO is operated in drop-out condition.	NMOS	–	95	130	µA	
		LV PMOS	–	80	150	µA	
		MV PMOS	–	85	120	µA	
Low-power mode							
Rated load current	Maximum load current at which all specifications are met. Higher load current is possible at increased headroom.	NMOS	30	–	–	mA	2
		LV PMOS	10	–	–	mA	
		MV PMOS	10	–	–	mA	
Load transient undershoot, overshoot	Load step of 0.5 × I_{RATED} in 1 µs, with baseline current of 0.01 × I_{RATED} or higher. Compared to final settled value. NMOS and LV PMOS: HR = 100 mV (PS1) MV PMOS: HR = 500 mV		–40	–	+70	mV	
Dropout voltage	From package V_{IN} pin to V_{OUT} pin. Load at LPM I_{RATED}.	NMOS	–	–	85	mV	
		LV PMOS	–	–	15	mV	
		MV PMOS	–	–	100	mV	
Overcurrent protection detection threshold (I_{OCP})	Threshold for OCP detection. OCP interrupt and self-shutdown are based on tripping this threshold.	NMOS	50	–	100	mA	
		LV PMOS	30	–	50	mA	
		MV PMOS	25	–	50	mA	
Ground current, no load	Measured at the battery. I_Q may much be higher if LDO is operated in dropout condition. PMOS UL excludes the FF/125 corner (~20 µA).	NMOS	–	20	23	µA	5
		LV PMOS	–	6.5	8	µA	
		MV PMOS	–	6.5	8	µA	
Normal and low-power mode							
Ground current, with load	% of the load current	NMOS	–	–	0.5	% I_{LOAD}	
		LV and MV PMOS	–	0.4	0.8	% I_{LOAD}	
LDO discharge time to below 100 mV	Strong pulldown enabled. Typical at nominal C_{OUT}, UL is with maximum C_{OUT}.	NMOS	–	0.1	2	ms	
		LV PMOS (except LV P600)	–	0.07	0.5	ms	
		LV P600	–	0.3	2.5	ms	
		MV PMOS (except MV P600)	–	–	2	ms	
		MV P600	–	–	3.5	ms	
VREG_OK threshold – during enable or step up			85%	90%	95%	V_{OUT}	

Table 3-17 LDO regulator specifications (cont.)

Parameter	Comments [1]	Min	Typ	Max	Units	Note
Forced bypass mode						
Ground current	NMOS	–	2	5	µA	6
	LV PMOS	–	0.75	1.5	µA	
	MV PMOS	–	0.5	1	µA	

1. All NMOS and LV PMOS specifications are specified with 100 mV (for PS1) and 160 mV (for PS2) headroom and measured at COUT, unless stated otherwise (like dropout voltage). All MV PMOS specifications are specified with 500 mV and measured at COUT unless stated otherwise (like dropout voltage).
2. The rated current is the current at which the regulator meets all specifications. Higher currents can be allowed, but the regulator may need more headroom, (this is, the difference between V_{IN} and V_{OUT}). For low-power mode, the LDO should be switched to normal power mode if the load current is expected to be above the rated current.
3. The settling time is for startup, and any upward voltage change with the rated load capacitance. Time is increased with larger load capacitance. Settling time of a downward voltage change depends on the load current.
4. Dropout voltage is defined as follows:
   a. Apply the specified load current.
   b. Set $V_{IN} = V_{OUT} + 0.5$ V.
   c. Measure the output voltage.
   d. Reduce V_{IN} until V_{OUT} is reduced by 100 mV.
   e. Calculate dropout voltage as $V_{IN} - V_{OUT}$ in this condition.
5. The low-power mode no-load ground current may be higher due to the current limiting mistrigger in the low headroom configuration. Disabling the current limiting feature or giving enough headroom per the dropout requirement can prevent mistriggering and reduce the ground current.
6. In bypass mode, there is an active gate to source clamp to protect the LV NMOS.

Table 3-18 L9_LVS specifications

Metric	Condition	Typical	Maximum	Units
Rdson		455	522	mΩ
Maximum load			30	mA
Maximum capacitance allowed	4.7 µF and 0.1 µF at SDR660 VDD_1P8 pin		4.8	µF

3.6.5 Internal voltage regulator connections

Some regulator supply voltages and/or outputs are connected internally to power other PMIC circuits. These circuits will not operate properly unless their source voltage regulators are enabled and set to their proper voltages. These requirements are summarized in Table 3-19.

Table 3-19 Internal voltage regulator connections

Voltage supply or regulator output	Default	Supported circuits
VPH_PWR	3.6 V	VIN0 for GPIO_06 to GPIO_09
VDD_IO	1.8 V	VIN1 for all GPIOs

3.7 GPIO specifications

Eight out of nine GPIO ports are digital I/Os that can be programmed for a variety of configurations (Table 3-20). Performance specifications for the different configurations are included in Section 3.4.

NOTE: All unused GPIO pins, except for the following, should be configured as digital inputs with 10 µA pull-down (their default state).

Table 3-20 Programmable GPIO configurations

Configuration type	Configuration description
Input	■ No pull-up ■ Pull-up (1.5 µA, 30 µA, or 31.5 µA) ■ Pull-down (10 µA) ■ Keeper
Output	■ Open-drain or CMOS ■ Inverted or noninverted ■ Programmable drive current

Most GPIOs default to digital input with 10 µA pull-down at power-on. Before they can be used for their intended purposes, they must be configured for use.

GPIOs are designed to run at a 4 MHz rate to support high-speed applications. The supported rate depends on the load capacitance and IR drop requirements. If the application specifies load capacitance, then the maximum rate is determined by the IR drop. If the application does not require a specific IR drop, then the maximum rate can be increased by increasing the supply voltage and adjusting the drive strength according to the actual load capacitance.

3.8 Power-on circuits and power sequence

Dedicated circuits continuously monitor several events that might trigger a power-on sequence. If any of these events occur, the PMIC circuits are powered on, the handset's available power sources are determined, the correct source is enabled, and the modem IC is taken out of reset.

The regulators that are included during the initial power-on sequence are determined by the hardware configuration control (OPTION pin), as defined in Section 3.10.

The sequence of signals detailed below support SM6125, followed by pertinent timing characteristics in Table 3-23.

Table 3-21 PM6125 power-on sequence for SM6125

Power on sequence [1]	Signal name	Vout primary (V)	Intended load
0	Power on trigger received		
1	VIO_OUT	1.8	PX0 PMIC IO pads
2	S6	1.352	MV sub regulation
3	S7	2.04	HV sub regulation
4	S5	0.928	MX, EBI_VDDIO, WCSS_MX, LPDDR4x_VDDQ (Subreg), WCSS_CX (Subreg)
5	S3_S4	0.872	CX + MSS
6	L8	0.728	VDD_WCSS_CX (0.73)
7	L9	1.8	LPDDR4x_VDD1(1.8), VDD_PX3,VDD_PX7, WCD (1.8 V),WCN (1.8 V), Sensors (1.8 V), RFFE (1.8 V), WSA (1.8 V)
7’	BOOST_BYP	3.5	External boost and bypass (if enabled by VREG_L9 supply rail)
8	VREF_MSM	1.25	VREF_HVPAD
9	S8	1.128	LV sub regulation, LPDDR4x_VDD2, SDR 1 V Digital (SubReg), VDDPX_1 (Subreg) 1.128 V (LPDDR4),1.224 V(LPDDR3) configured by option pin
10	L3	1	SDR(D1V)
11	VREG_L9_LVS	1.8	SDR 1.8 V IO supply via MOS switch (only enabled for SM6125 platform)
12	L6	0.624	LPDDR4x_VDDQ (0.6), VDDIO_EBI, VDDIO_CK_EBI
13	L11	1.8	EMMC1_VDDQ (1.8),UFS (1.8)
14	VREF_LPDDR3	0.612	LPDDR3 VREF (only enabled for SM6125 if configured for LPDDR3. See option pin configurations, VDD_L2_L3_L4/2)
15	L18	1.232	EBI_PLL(1.23), MIPI_CSI, MIPI_DSI (1.23), VDDPX_10
16	L7	0.928	USB 0.9 V
17	L10	1.8	USB2 (1.8V), USB3.1 (1.8), QFPROM (1.8V), UFS (1.8), VDDPX_11
18	L4	0.928	EBIPLL (0.93 V), USB2(A0P9), USB3(A0P9),QLINK,QREF, MIPI_DSI,UFS(0.9)
19	L15	3.128	USB3(3.1)
20	L12	1.8	Camera_sensors (1.8),USB_Redriver, MIPI_VDDIO (1.8)

Table 3-21 PM6125 power-on sequence for SM6125 (cont.)

Power on sequence [1]	Signal name	Vout primary (V)	Intended load
21	L24	2.96	EMMC1_VCC(2.96)/UFS_MEM(2.96)
22	L5	2.96	PX2 (PON dependent on Option pin decode. Power ON only if SD_CARD_BOOT selected)
23	L22	2.96	MEM_SD_MMC_VDD(2.96) (PON dependent on Option pin decode. Power ON only if SD_CARD_BOOT is selected)
24	S1_S2	0.872	APC0, APC1
25			PON_RESET_N release
	PON done		ACK PON complete

1. The power-off sequence is the inverse order of the power-on sequence. The LNBBCLK supply rail powers off after the clock request is turned off in the power-off sequence.

Table 3-22 PM4250 power-on sequence for SM4250/SM6115

Power on sequence [1]	Signal name	Vout primary (V)	Intended load
0	Power on trigger received		
1	VIO_OUT	1.8	PX0 PMIC IO pads
2	S6A	1.352	MV sub regulation
3	S7A	2.04	HV sub regulation
4	S5A	0.91	MX, EBI_VDDIO, WCSS_MX, LPDDR4x_VDDQ (Subreg), WCSS_CX (Subreg)
5	L2A	0.928	LPI_MX
6	S3A_S4A	0.87	CX + MSS
7	L3A	0.87	LPI_CX
8	L8A	0.73	VDD_WCSS_CX (0.73)
9	L9A	1.8	LPDDR4x_VDD1 (1.8), VDD_PX3, VDD_PX7, WCD (1.8 V),WCN (1.8 V), Sensors (1.8 V), RFFE (1.8 V), WSA (1.8 V)
10	L13A	1.8	WTR 1.8 V
11	L1A	1	WTR 1 V
12	L7A	1.256	UIM/SDC VBIAS
13	L18A	1.232	EBI_PLL (1.23), MIPI_CSI, MIPI_DSI (1.23), VDDPX_10
14	eLDO_9	0.928	(Optional) USB SS Core
15	L4A	0.928	EBIPLL (0.93 V), USB2 (A0P9), USB3 (A0P9), MIPI_DSI, UFS (0.9), USB 0.9 V
16	S8A	1.128	LV sub regulation, LPDDR4x_VDD2, VDDPX_1
17	L6A	0.624	LPDDR4x_VDDQ (0.6), VDDIO_EBI, VDDIO_CK_EBI

Table 3-22 PM4250 power-on sequence for SM4250/SM6115 (cont.)

Power on sequence [1]	Signal name	Vout primary (V)	Intended load
18	L12A	1.8	WCN (1.8 V), WCSS_PLL, MIPI_VDDIO (1.8)
19	L10A	1.8	BBCLK_VREG
20	L15A	3.128	USB3 (3.1)
21	L11A	1.8	EMMC1_VDDQ (1.8), UFS (1.8)
22	L24A	2.96	EMMC1_VCC (2.96)/UFS_MEM (2.96)
23	L5A	2.96	PX2
24	L22A	2.96	MEM_SD_MMC_VDD (2.96)
25	S1A_S2A	0.8	APC0, APC1
	PON done		ACK PON complete

1. The power-off sequence is the inverse order of the power-on sequence. The LNBBCLK supply rail powers off after the clock request is turned off in the power-off sequence.

Table 3-23 Power-on timing specifications

Parameter [1]	Minimum	Typical	Maximum	Units
t_{reg1} – power-on event to first regulator on [2]	26	–	200	ms
t_{reset0} – PON_RESET_N low to first regulator off	–	10	–	ms
t_{reg} – inter-regulator turn-on time	–	200	–	µs
t_{off} – inter-regulator turn-off time	–	1000	–	µs
t_{ps_hold} – PS_HOLD time out [3]	150	200	250	ms

1. treg0 and tsettle specifications are covered by the individual module (SMPS, LDO) specifications.
2. The first-regulator power-on time (treg1) depends on the bandgap-reference decoupling capacitor at REF_ BYP. The specified value is based on 0.1 µF. If these debounce timers are increased, then the treg1 value will also increase.
3. PS_HOLD timeout is the time after which the PMIC turns off if PS_HOLD is not yet driven high enough by the SDM device.

3.9 OPT hardware controls

3.9.1 PM6125/PM4250 OPTION hardware control

The OPTION pin must be hardwired to ground or VREG_IO/VPH_PWR, or be left open (Hi-Z); Table 3-24 lists the specific configurations based on the termination of the OPTION pin.

Table 3-24 PM6125 hardware configuration options

Option pin	Configuration
5.6 K	No external boost bypass, SD card boot, LPDDR3
22 K	No external boost bypass, SD card boot, LPDDR4X

Table 3-24 PM6125 hardware configuration options

Option pin	Configuration
47 K	No external boost bypass, no SD card boot, LPDDR3
75 K	No external boost bypass, no SD card boot, LPDDR4X
130 K	External boost bypass, SD card boot, LPDDR3
240 K	External boost bypass, SD card boot, LPDDR4X
330 K	External boost bypass, no SD card boot, LPDDR3
750 K	External boost bypass, no SD card boot, LPDDR4X

NOTE: SD card boot implies the voltages for SD card will be enabled during power on sequence. No SD card boot implies the voltages for SD card will be disabled during power on sequence.
For PM4250, external boost bypass options do not apply.

3.10 Undervoltage lockout

The handset supply voltage (VPH_PWR) is monitored continuously by a circuit that automatically turns off the device at severely low VPH conditions.

To power on successfully, VPH must be between the UVLO rising threshold and the OVLO falling threshold. These thresholds are not used after PON. However, in order to stay on, VPH must remain between the UVLO falling threshold and the OVLO rising threshold. If these thresholds are exceeded, a fault will occur and the IC will immediately shut down.

Other than the programmable threshold, software is not involved in UVLO/OVLO detection. Hysteresis and time delays are not programmable, and UVLO/OVLO events do not generate interrupts. They are reported to the modem IC via the PON_RESET_N signal. UVLO/OVLO-related voltage and timing specifications are listed in Table 3-25.

Table 3-25 UVLO and OVLO thresholds

Parameter	Minimum	Maximum	Hardware default	SBL setting	Units
UVLO falling threshold	1.50	3.05	2.40	2.00	V
UVLO rising threshold	1.65	3.65	2.85	2.60	V
UVLO interval detection	–	–	3	–	µs
OVLO rising threshold	4.20	7.30	5.90	5.50	V
OVLO falling threshold	2.20	7.05	5.40	5.00	V
OVLO detection interval	–	–	6	–	µs

3.11 General housekeeping

The PMIC includes many circuits that support handset-level housekeeping functions – various tasks that must be performed to keep the handset in order. Integration of these functions reduces the external parts count and the associated size and cost. Housekeeping functions include an analog switch matrix, multiplexers, and voltage scaling; an HK/XO ADC circuit; system clock circuits; a real-time clock for time and alarm functions; and overtemperature protection.

3.11.1 Analog multiplexer and scaling circuits

A set of analog switches, analog multiplexers, and voltage-scaling circuits select and condition a single analog signal for routing to the on-chip HK/XO ADC. The multiplexer and scaling functions are summarized in Table 3-26.

Table 3-26 Analog multiplexer and scaling functions

Channel number (hex.)	Description	Source	Scaling	Internal pull-up	Input range (V)
0	REF_GND	Pin: REF_GND	1/1	Open	0 to 1.875
1	1.25VREF	Internal: MBG	1/1	Open	0 to 1.875
2	VREF_VADC	Internal: VADC LDO	1/1	Open	0 to 1.875
82	VREF_VADC	Internal: VADC LDO	1/1	Open	0 to 1.875
83	VPH_PWR	Pin: VPH_PWR	1/3	Open	0 to 5
84	VBAT_SNS	Pin: VBAT_SNS	1/3	Open	0 to 5
85	VCOIN	Pin: VCOIN	1/3	Open	0 to 3.6
6	DIE_TEMP	Internal: TEMP_ALARM	1/1	Open	0 to 1.875
C	XO_THERM	Pin: XO_THERM	1/1	Open	0 to 1.875
D	AMUX_THM1	Pin: AMUX_1	1/1	Open	0 to 1.875
E	AMUX_THM2	Pin: AMUX_2	1/1	Open	0 to 1.875
12	AMUX1_GPIO	Pin: GPIO3 (LV)	1/1	Open	0 to 1.875
13	AMUX2_GPIO	Pin: GPIO4 (LV)	1/1	Open	0 to 1.875
14	AMUX3_GPIO	Pin: GPIO6 (MV)	1/1	Open	0 to 1.875
15	AMUX4_GPIO	Pin: GPIO7 (MV)	1/1	Open	0 to 1.875
2C	XO_THERM	Pin: XO_THERM	1/1	30 k	0 to 1.875
2D	AMUX_THM1	Pin: AMUX_1	1/1	30 k	0 to 1.875
2E	AMUX_THM2	Pin: AMUX_2	1/1	30 k	0 to 1.875
32	AMUX1_GPIO	Pin: GPIO3 (LV)	1/1	30 k	0 to 1.875
33	AMUX2_GPIO	Pin: GPIO4 (LV)	1/1	30 k	0 to 1.875
34	AMUX3_GPIO	Pin: GPIO6 (MV)	1/1	30 k	0 to 1.875
35	AMUX4_GPIO	Pin: GPIO7 (MV)	1/1	30 k	0 to 1.875
4C	XO_THERM	Pin: XO_THERM	1/1	100 k	0 to 1.875

Table 3-26 Analog multiplexer and scaling functions (cont.)

Channel number (hex.)	Description	Source	Scaling	Internal pull-up	Input range (V)
4D	AMUX_THM1	Pin: AMUX_1	1/1	100 k	0 to 1.875
4E	AMUX_THM2	Pin: AMUX_2	1/1	100 k	0 to 1.875
52	AMUX1_GPIO	Pin: GPIO3 (LV)	1/1	100 k	0 to 1.875
53	AMUX2_GPIO	Pin: GPIO4 (LV)	1/1	100 k	0 to 1.875
54	AMUX3_GPIO	Pin: GPIO6 (MV)	1/1	100 k	0 to 1.875
55	AMUX4_GPIO	Pin: GPIO7 (MV)	1/1	100 k	0 to 1.875
6C	XO_THERM	Pin: XO_THERM	1/1	400 k	0 to 1.875
6D	AMUX_THM1	Pin: AMUX_1	1/1	400 k	0 to 1.875
6E	AMUX_THM2	Pin: AMUX_2	1/1	400 k	0 to 1.875
72	AMUX1_GPIO	Pin: GPIO3 (LV)	1/1	400 k	0 to 1.875
73	AMUX2_GPIO	Pin: GPIO4 (LV)	1/1	400 k	0 to 1.875
74	AMUX3_GPIO	Pin: GPIO6 (MV)	1/1	400 k	0 to 1.875
75	AMUX4_GPIO	Pin: GPIO7 (MV)	1/1	400 k	0 to 1.875

3.11.2 HK/XO ADC circuit

The analog-to-digital converter circuit is shared by the housekeeping (HK) and 38.4 MHz crystal oscillator (XO) functions.

HK/XO ADC performance specifications are listed in Table 3-27.

Table 3-27 VADC electrical specification

Specification	Test condition	Min	Typ	Max	Units
1/1 channel end-to-end accuracy	Calibrated data result	-11	±6	11	mV
1/1 channel end-to-end accuracy with internal pull-up	Calibrated data result	-12.5	±7	12.5	mV
1/3 channel end-to-end accuracy	Calibrated data result	-20	±10	20	mV
ADC resolution (LSB)	1/1 channel Scaled to 1/3 channel	– –	64.879 194.637	– –	µV
Analog bandwidth (anti-alias filter)		–	500	–	kHz
ADC LDO voltage		1.828	1.875	1.910	V
ADC sample clock		–	4.8	–	MHz
ADC conversion time	1 K decimation ratio, 4.8 MHz sample clock	–	654	700	µs
Current consumption	VADC active	–	450	500	µA
100 K pull-up	Trimmed value	99.5	100	100.5	kΩ

Table 3-27 VADC electrical specification (cont.)

Specification	Test condition	Min	Typ	Max	Units
400 K pull-up	Trimmed value	398	400	402	kΩ
30 K pull-up	Trimmed value	29.7	30	30.3	kΩ
Pull-up temperature coefficient	For 400 K pull-up resistor	-100	–	100	ppm/°C
	For 100 K and 30 K pull-up resistor	-225	–	225	ppm/°C
1/1 channel AMUX input resistance		10	–	–	MΩ
1/3 channel AMUX input resistance		1	–	–	MΩ

3.11.3 System clocks

The PMIC includes several clock circuits whose outputs are used for general housekeeping functions and elsewhere within the handset system. These circuits include a 38.4 MHz XO with multiple controllers and buffers, an MP3 clock output, an RC oscillator, and sleep-clock outputs. Performance specifications for these functions are presented in the following subsections.

3.11.3.1 38.4 MHz XO circuits

An external crystal is supplemented by on-chip circuits to generate the intended 38.4 MHz reference signal. Using an external thermistor network, the on-chip ADC, and advanced temperature compensation software, the PMIC eliminates the large and expensive VCTCXO module required by previous-generation chipsets. The XO circuits initialize and maintain valid pulse waveforms and measure time intervals for higher-level handset functions.

Multiple controllers manage the XO and signal buffering and generate the intended clock outputs (all derived from one source):

- RF_CLKx and LN_BB_CLKx low-noise outputs – enabled internally through SPMI write (RF_CLKx is 38.4 MHz while LN_BB_CLKx is 19.2 MHz).

Since the different controllers and outputs are independent, circuits other than those needed for the WAN can operate even while the modem IC is asleep and its RF circuits are powered down.

The XTAL_IN and XTAL_OUT pins are incapable of driving a load – the oscillator will be significantly disrupted if either pin is externally loaded.

The XTAL_IN pin can optionally work as a clock input from PMK8002 clock IC.

The 38.4 MHz XO circuit and related performance specifications are listed in the following tables.

Table 3-28 RF_CLKx specification

Parameter	Comments	Min	Typ	Max	Unit
Frequency	Set by external crystal	–	38.4	–	MHz
Duty cycle	After running duty cycle calibration	49.5	50	50.5	%
Output voltage swing		1.3	1.4	1.5	V
Output buffer impedance	1x	–	39	–	Ω
	2x	–	30	–	Ω
	3x	–	20	–	Ω
	4x	–	11	–	Ω
Phase noise	At 1 kHz	–	–	-130	dBc/Hz
	At 10 Hz	–	–	-80	dBc/Hz
	At 10 kHz	–	–	-142	dBc/Hz
	At 100 Hz	–	–	-104	dBc/Hz
	At 100 kHz	–	–	-154	dBc/Hz
	At 1 MHz	–	–	-154	dBc/Hz
AM noise	At 1 kHz	–	–	-87	dBc/Hz
	At 10 kHz	–	–	-122	dBc/Hz
	At 100 kHz	–	–	-137	dBc/Hz
	At 1 MHz	–	–	-137	dBc/Hz

Table 3-29 LN_BB_CLKx specification

Parameter	Comments	Min	Typ	Max	Unit
Frequency	Set by the external crystal	–	19.2	–	MHz
Duty cycle		48	50	52	%
Output voltage swing		1.746	1.8	1.854	V
Output buffer impedance	1x	–	46	–	Ω
	2x	–	29	–	Ω
	3x	–	19	–	Ω
	4x	–	12	–	Ω

Table 3-29 LN_BB_CLKx specification

Parameter	Comments	Min	Typ	Max	Unit
Phase noise	At 10 Hz	–	–	-86	dBc/Hz
	At 100 Hz	–	–	-110	dBc/Hz
	At 1 kHz	–	–	-136	dBc/Hz
	At 10 kHz	–	–	-144	dBc/Hz
	At 100 kHz	–	–	-144	dBc/Hz
	At 1 MHz	–	–	-144	dBc/Hz
Period jitter	RMS	–	–	3	ps

Table 3-30 Divided-down XO clock output specifications

Parameter	Min	Typ	Max	Units
Buffer output impedance				
at low GPIO drive strength	–	42	–	Ω
at medium GPIO drive strength	–	30	–	Ω
at high GPIO drive strength	–	22	–	Ω
Phase noise				
at 100 Hz	–	-85	–	dBc/Hz
at 1 kHz	–	-95	–	dBc/Hz
at 10 kHz	–	-100	–	dBc/Hz
at 100 kHz	–	-105	–	dBc/Hz
at 250 kHz	–	-105	–	dBc/Hz
at 500 kHz	–	-105	–	dBc/Hz

3.11.3.2 38.4 MHz XO crystal requirements

Crystal performance is critical to a wireless product's overall performance. Guidance is available within the *38.4 MHz Modem Crystal Qualification Requirements and Approved Suppliers* (80-NJ458-19) document. This document includes:

- Data needed from crystal suppliers to demonstrate compliance
- Approved suppliers for different crystal configurations
- Description of various schematic options

3.11.3.3 Sleep clock

The sleep clock is generated by:

- Using the 38.4 MHz XO circuit and dividing its output by 1172 to create a 32.7645 kHz signal.

The PMIC sleep-clock output is routed to the modem IC via SLEEP_CLK. It is also available for other applications using properly configured GPIOs. Table 3-31 shows the sleep clock performance specifications.

Table 3-31 Sleep clock performance specifications

Parameter	Comments	Min	Typ	Max	Units
Period jitter (RMS)	XO/1172 source; as defined in JDSE6	–	10	–	ns
Duty cycle	XO/1172 source	–	50	–	%
Tolerance	XO/1172 source	-20	–	+20	ppm

Related specifications presented elsewhere include:

- 38.4 MHz XO circuits (Section 3.11.3.1)
- Output characteristics (voltage levels, drive strength, etc.), as defined in Section 3.4

3.11.4 Real-time clock

The real-time clock (RTC) functions are implemented by a 32-bit real-time counter and one 32-bit alarm; both are configurable in one-second increments. The primary input to the RTC circuits is the selected sleep-clock source (calibrated low-frequency oscillator, or divided-down 38.4 MHz XO). Even when the phone is off, the selected oscillator and RTC continue to run off the main battery.

If only the main battery is present and an SMPL event occurs, then RTC contents are corrupted. The phone must reacquire system time from the network to resume the usual RTC accuracy. Similarly, if the main battery is not present and the voltage at VCOIN drops too much, RTC contents are again corrupted. In either case, the RTC reset interrupt is generated. A different interrupt is generated if the oscillator stops, also causing RTC errors.

If RTC support is needed when the battery is removed, a qualified coin-cell or super capacitor is required on the VCOIN pin of the PMIC. If only SMPL support is needed when battery is removed, a 47 µF capacitor with at least 10 µF effective capacitance at 3 V is required on the VCOIN pin of the PMIC.

Pertinent RTC specifications are listed in Table 3-32.

Table 3-32 RTC performance specifications

Parameter	Comments	Min	Typ	Max	Units
Tuning resolution	With known calibrated source	–	3.05	–	ppm
Tuning range		-192	–	+192	ppm
Accuracy XO/1172 as RTC source CalRC as RTC source	 Phone on Phone off, valid battery present Phone off, valid coin cell present [1]	 – – –	 – – –	 24 50 200	 ppm ppm ppm

1. Assumes a maximum ESR of coin cell/super capacitor is 1 kΩ. For a maximum ESR of 2 kΩ, the accuracy is 500 ppm.

3.11.5 Overtemperature protection (smart thermal control)

The PMIC includes overtemperature protection in stages, depending on the level of urgency as the die temperature rises:

- Stage 0: normal operating conditions (less than 95°C).
- Stage 1: 95–115°C; an interrupt is sent to the modem IC.
- Stage 2: 115–145°C; an interrupt is sent to the modem IC.
- Stage 3 – greater than 145°C; an interrupt is sent to the modem IC, and the PMIC is completely shut down.

Temperature hysteresis is incorporated, such that the die temperature must cool significantly before the device can be powered on again. If any start signals are present while at Stage 3, they are ignored until Stage 0 is reached. When the device cools enough to reach Stage 0 and a start signal is present, the PMIC will power up immediately.

4 Mechanical information

4.1 Device physical dimensions

The PM6125/PM4250 is available in the FOWNSP144 that includes dedicated ground pins for improved grounding, mechanical strength, and thermal continuity. The FOWNSP144 has a 4.81 mm by 4.81 mm body, with a maximum height of 0.57 mm. Pin 1 is located by an indicator mark on the top of the package, and by the ball pattern when viewed from below. A simplified version of the FOWNSP144 outline drawing is shown in Figure 4-1.

NOTE: Click the following link to download the *Package Outline Drawing, FOWNSP144, 4.81 × 4.81 × 0.57 mm, M410* (NT90-PK502-1) from the Qualcomm® CreatePoint website.

https://createpoint.qti.qualcomm.com/search/contentdocument/stream/dcn/NT90-PK502-1

After successfully logging in, the document is downloaded.

NOTE: Make this document a favorite to be notified of any changes.

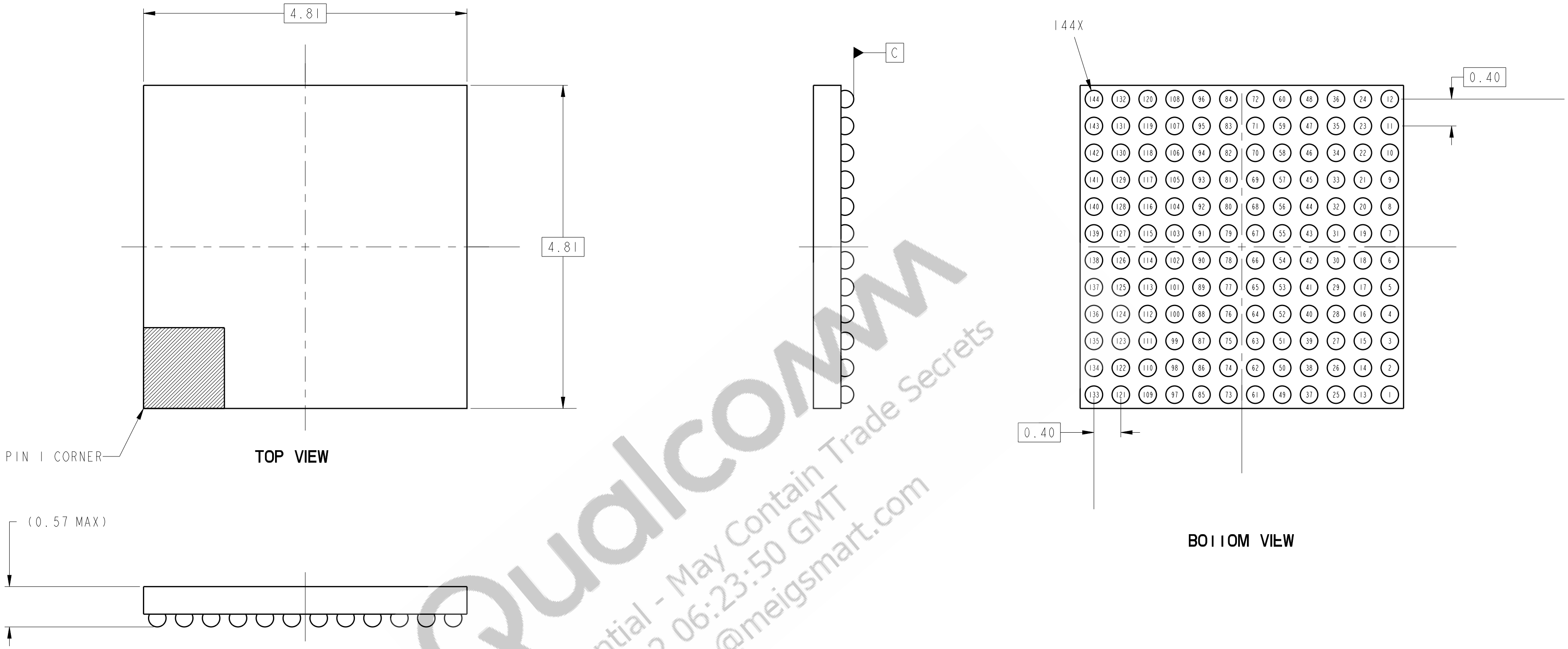


Figure 4-1 FOWNSP144 outline drawing

NOTE: This is a simplified outline drawing. Click the link on the previous page to download the complete, up-to-date package outline drawing.

4.2 Part marking

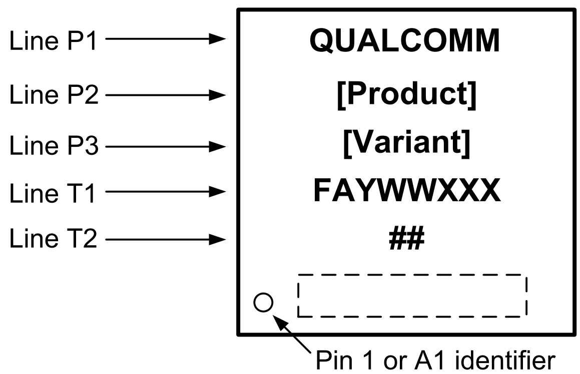


Figure 4-2 PM6125/PM4250 device marking (top view, not to scale)

Table 4-1 PM6125/PM4250 device marking line definitions

Line	Marking	Description
P1	QUALCOMM	Qualcomm name
P2	[Product]	Qualcomm Technologies, Inc. (QTI) product name ■ PM6125 ■ PM4250
P3	[Variant] (PRR-BB)	Device variant information ■ See Table 4-3 for the assigned values.
T1	FAYWWXXX	F = wafer fab source of supply code ■ F = F for TSMC ■ F = P for SMIC ■ F = J for Samsung A = assembly site (ball drop) code ■ A = E for ASE, Taiwan ■ A = Z for Amkor Technologies, Portugal Y = single/last digit of year WW = two-digit work week of year specified by Y XXX = sequential serial number portion of lot
T2	##	## = wafer number
–	○ Blank or random	Pin 1 or pin A1 indicator Additional content, as necessary

4.3 Device ordering information

4.3.1 Specification-compliant devices

The Oracle short description is used to order QTI products, and is present on both the customer label and this document. The short description includes the product name, configuration code, package type, product revision code, source code, and feature code/program ID of the part.

This device can be ordered using the identification code shown in Table 4-2.

Table 4-2 Device identification code

Device ID code	AAA-AAAA	-P	-TTTTTT	NNNN	A	+FF	-EE	-RR	-S	-BB or -PID [1]
Symbol definition	Product name	Configuration code	Package type	Number of pins	Package variable	Additional package information	Shipping package	Product revision	Source code	Feature code
Example 1	PM-6125	-0	-FOWNSP	144			-TR	-01	-0	
Example 2	PM-4250	-0	-FOWNSP	144			-TR	-00	-0	

1. The feature code (BB) and the program ID (PID) are mutually exclusive. A product may have one of them or none of them, but it will never have both. If there is no feature code/program ID, this field is blank, and the Oracle short description ends after the source configuration code (S).

For example:

- Example 1: PM-6125-0-FOWNSP144-TR-01-0
- Example 2: PM-4250-0-FOWNSP144-TR-00-0

Table 4-2 shows the current package-type nomenclature. For legacy parts, the Oracle short description has the position of package type and number of pins reversed.

Device identification details for all samples available to date are summarized in Table 4-3.

Table 4-3 Device identification details

Device	Sample type	Product configuration code (P)	Product revision code (RR)	Feature code (BB) [1]	Hardware version	Source configuration code (S) [2]
PM6125	ES1	0	00	–	v1.0	0
PM6125	ES2/CS [3]	0	01	–	v1.1	0 or 1
PM4250	ES/CS [4]	0	00	–	v1.1	0

1. BB is the feature code that identifies an IC's specific feature set, which distinguishes it from other versions or variants.
2. S is the source configuration code that identifies all the qualified die fabrication-source combinations available when the particular sample type was shipped. S values are defined in Table 4-4.
3. PM6125 ES2 and CS parts have the same PRR code. All devices with the date code YWW ≥ 912 are CS quality.
4. PM4250 ES and CS parts have the same PRR code. All devices with the date code YWW ≥ 003 are CS quality.

Table 4-4 Source configuration code

S value	Die	F value = F	F value = P	F value = J
0	Digital	TSMC	SMIC	–
1	Digital	TSMC	SMIC	Samsung

4.3.2 Daisy chain devices

The PM6125/PM4250 daisy chain ordering part number is TBD.

4.4 Device moisture-sensitivity level

Plastic-encapsulated surface mount packages are susceptible to damage induced by absorbed moisture and high temperature. A package's moisture-sensitivity level (MSL) indicates its ability to withstand exposure after it is removed from its shipment bag, while it is on the factory floor awaiting PCB installation. A low MSL rating is better than a high rating; a low MSL device can be exposed on the factory floor longer than a high MSL device. All pertinent MSL ratings are summarized in Table 4-5.

Table 4-5 MSL ratings summary

MSL	Out-of-bag floor life	Comments
1	Unlimited	≤ 30°C/85% RH
2	1 year	≤ 30°C/60% RH
2a	4 weeks	≤ 30°C/60% RH
3	168 hrs.	≤ 30°C/60% RH; PM6125/PM4250 rating
4	72 hrs.	≤ 30°C/60% RH
5	48 hrs.	≤ 30°C/60% RH
5a	24 hrs.	≤ 30°C/60% RH
6	Mandatory bake before use. After bake, must be reflowed within the time limit specified on the label.	≤ 30°C/60% RH

QTI follows the latest IPC/JEDEC J-STD-020 standard revision for moisture-sensitivity qualification. ***The PM6125/PM4250 devices are classified as MSL3; the qualification temperature was 260°C +0°/-5°C***. This qualification temperature (260°C +0°/-5°C) should not be confused with the peak temperature within the recommended solder reflow profile.

4.5 Thermal characteristics

Rather than provide thermal resistance values θ_{JC} and θ_{JA}, validated thermal package models are provided through the CreatePoint website. Designers can extract thermal resistance values by conducting their own thermal simulations.

NOTE: Click the following link to download the *PM6125 Package Thermal Model Icepak* (HS11-VV033-5HW) from the CreatePoint website.

https://createpoint.qti.qualcomm.com/search/contentdocument/stream/dcn/HS11-VV033-5HW

Click the following link to download the *PM6125 Package Thermal Model Flowtherm* (HS11-VV033-6HW) from the CreatePoint website.

https://createpoint.qti.qualcomm.com/search/contentdocument/stream/dcn/HS11-VV033-6HW

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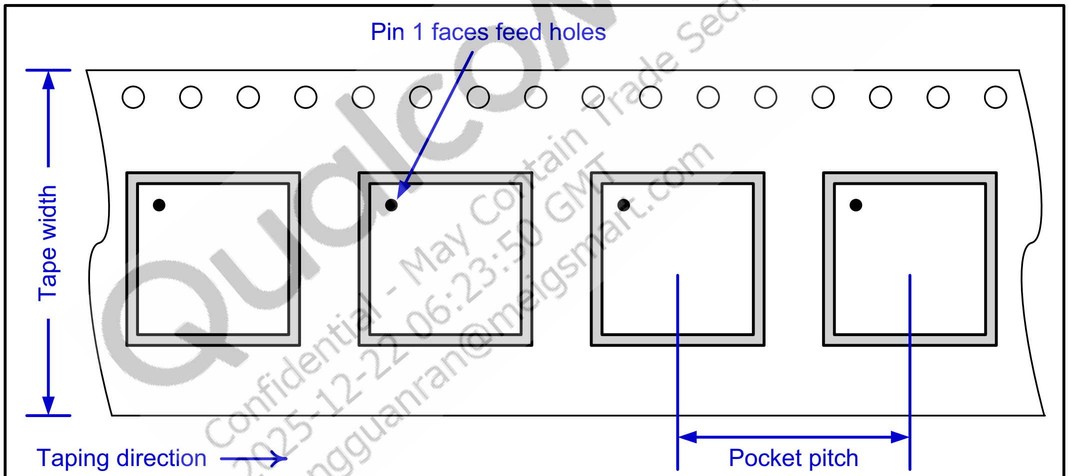
5 Carrier, storage, and handling information

5.1 Carrier

5.1.1 Tape and reel information

All QTI carrier tape systems conform to EIA-481 standards.

A simplified sketch of the PM6125/PM4250 tape carrier is shown in Figure 5-1, including the proper part orientation, maximum number of devices per reel, and key dimensions.



Tape feed: Single	Reel diameter: 330 mm	Tape width: 12 mm
Units per reel: 4000	Hub diameter: 178 mm	Pocket pitch: 8 mm

Figure 5-1 Carrier tape drawing with part orientation

Tape-handling recommendations are shown in Figure 5-2.

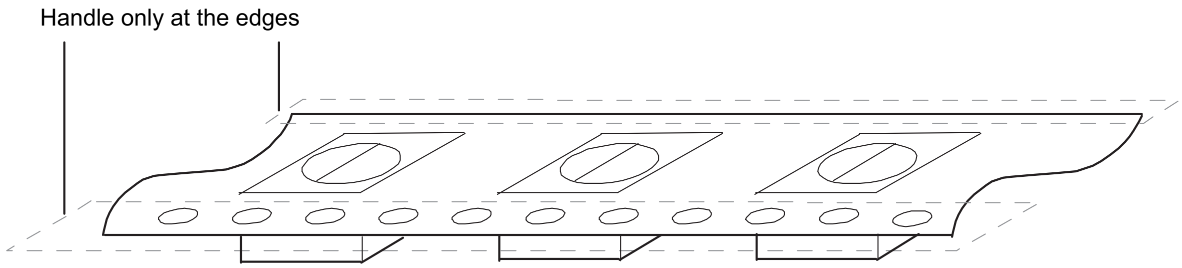


Figure 5-2 Tape handling

5.2 Storage

5.2.1 Bagged storage conditions

PM6125/PM4250 devices delivered in tape and reel carriers must be stored in sealed, moisture barrier, antistatic bags. See the *IC Products Packing Method* (80-VK055-1) for the expected shelf life.

5.2.2 Out-of-bag duration

The out-of-bag duration is the time a device can be on the factory floor before being installed onto a PCB. It is defined by the device MSL rating, as described in Section 4.4.

5.3 Handling

Tape handling was described in Section 5.1.1. Other (IC-specific) handling guidelines are presented below.

5.3.1 Baking

It is not necessary to bake the PM6125 device if the conditions specified in Section 5.2.1 and Section 5.2.2 have not been exceeded.

It is necessary to bake the PM6125 device if any condition specified in Section 5.2.1 or Section 5.2.2 has been exceeded. The baking conditions are specified on the moisture-sensitive caution label attached to each bag; see the *IC Products Packing Method* (80-VK055-1) document for details.

CAUTION: If baking is required, the devices must be transferred into trays that can be baked to at least 125°C. Devices should not be baked in tape and reel carriers at any temperature.

5.3.2 Electrostatic discharge

Electrostatic discharge (ESD) occurs naturally in laboratory and factory environments. An established high-voltage potential is always at risk of discharging to a lower potential. If this discharge path is through a semiconductor device, destructive damage may result.

ESD countermeasures and handling methods must be developed and used to control the factory environment at each manufacturing site.

QTI products must be handled according to the ESD Association standard: ANSI/ESD S20.20-1999, *Protection of Electrical and Electronic Parts, Assemblies, and Equipment*.

5.4 Bar code label and packing for shipment

See the *IC Products Packing Method* (80-VK055-1) for all packing-related information, including bar code label details.

6 PCB mounting guidelines

6.1 RoHS compliance

The device complies with the requirements of the EU RoHS directive. Its SnAgCu solder balls use SAC405 composition. A product material declaration (PMD) that provides RoHS and other product environmental governance information is published when the data is available.

6.2 SMT assembly guidelines

For recommendations on SMT process development, see the *SMT Assembly Guidelines* (SM80-P0982-1).

NOTE: Click the following link to download the *SMT Assembly Guidelines* (SM80-P0982-1) from the CreatePoint website.

https://createpoint.qti.qualcomm.com/search/contentdocument/stream/dcn/SM80-P0982-1

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6.3 Daisy chain components

Daisy chain packages use the same processes and materials as actual products; they are recommended for SMT characterization and board-level reliability testing. The SMT process recommendations described in Section 6.2 can be performed using daisy chain components.

Ordering information is given in Section 4.3.2.

Daisy chain PCB routing recommendations are available for download.

NOTE: Click the following link to download the *Daisy Chain Interconnect, 144FOWNSP, 4.81 × 4.81 × 0.57 mm* (TBD) from the CreatePoint website.

This link will be provided in a future revision of this document.

After successfully logging in, the document is downloaded.

NOTE: Make this document a favorite to be notified of any changes.

7 Part reliability

7.1 Reliability qualifications summary: TSMC and SMIC

Table 7-1 Silicon reliability results: TSMC and SMIC

Device qualification Tests, standards, and conditions	Sample size SMIC	Sample size TSMC	Sample size SEC	Result
Early life failure rate (ELFR) in DPPM HTOL: JESD22-A108-A Total samples from three different wafer lots	160	80	231 (77 × 3 fab lots)	Pass DPPM < 1000 [1]
Average failure rate (AFR) in FIT (λ) failures per billion device hours; functional HTOL JESD22-A108-A (Total samples from three different wafer lots)	160	80	231 (77 × 3 fab lots)	Pass FIT < 100 [1]
Mean time to failure (MTTF) t = 1/λ in million hours Total samples from three different wafer lots	160	80	231 (77 × 3 fab lots)	Pass MTTF > 20
ESD – human body model (HBM) rating JESD22-A114-F	84	42	3 (1 fab lot)	Pass 2000 V
ESD – charged device model (CDM) rating JESD22-C101-E	24	12	3 (1 fab lot)	Pass 500 V
Latch-up (I-test): EIA/JESD78A Trigger current: ±100 mA; temperature: 85°C	24	12	3 (1 fab lot)	Pass
Latch-up (Vsupply overvoltage): EIA/JESD78A Trigger voltage: Each VDD pin, stress at 1.5 × Vddmax per device specification; temperature: 85°C	24	12	3 (1 fab lot)	Pass

1. The cumulative DPPM/FITs from multiple products were under the foundry TSMC/SMIC, PMIC5 150 nm

Table 7-2 Package reliability results for SMIC

Package qualification Tests, standards, and conditions	Sample size	ASE sample size	ATEP sample size
Moisture resistance test (MRT): MSL3; J-STD-020- D 3 × reflow cycles at 255°C +5/-0°C	462	Pass	Pass
Temperature cycle: JESD22-A104-D Temperature: -55°C to 125°C; number of cycles: 1000 Minimum soak time at minimum/maximum temperature: 5 minutes Cycle rate: 2 cycles per hour (CPH) MSL3 preconditioning: JESD22-A113-F Reflow temperature: 255°C +5/-0°C	231	Pass	Pass
Unbiased highly accelerated stress test (UHAST): JESD22-A118 MSL3 preconditioning: JESD22-A113-F Reflow temperature: 255°C +5/-0°C	231	Pass	Pass
High-temperature storage life: JESD22-A103-D Temperature 150°C, 1000 hours	231	Pass	Pass
Physical dimensions: JESD22-B100-A	15	Pass	Pass
Solder ball shear: JESD22-B117-B Total samples from three different assembly lots at each SAT	45	Pass	Pass

Table 7-3 Package reliability results for TSMC

Package qualification Tests, standards, and conditions	Sample size	ASE sample size	ATEP sample size
Moisture resistance test (MRT): MSL3; J-STD-020- D 3 × reflow cycles at 255°C +5/-0°C	462	Pass	Pass
Temperature cycle: JESD22-A104-D Temperature: -55°C to 125°C; number of cycles: 1000 Minimum soak time at minimum/maximum temperature: 5 minutes Cycle rate: 2 cycles per hour (CPH) MSL3 preconditioning: JESD22-A113-F Reflow temperature: 255°C +5/-0°C	231	Pass	Pass
Unbiased highly accelerated stress test (UHAST): JESD22-A118 MSL3 preconditioning: JESD22-A113-F Reflow temperature: 255°C +5/-0°C	231	Pass	Pass
High-temperature storage life: JESD22-A103-D Temperature 150°C, 1000 hours	231	Pass	Pass

Table 7-3 Package reliability results for TSMC

Package qualification Tests, standards, and conditions	Sample size	ASE sample size	ATEP sample size
Physical dimensions: JESD22-B100-A	15	Pass	Pass
Solder ball shear: JESD22-B117-B Total samples from three different assembly lots at each SAT	45	Pass	Pass

Table 7-4 Package reliability results for SEC

Package qualification Tests, standards, and conditions	Sample size	ASE sample size	Amkor sample size
Moisture resistance test (MRT): MSL 1; J-STD-020-D 3x reflow cycles at 260 +0/-5°C	693 (231 × 3 assembly lots per SAT)	Pass	Pass
Temperature cycle: JESD22-A104-D Temperature: -55 to +125°C; number of cycles:1000 Minimum soak time at minimum/maximum temperature: five minutes Cycle rate: two cycles per hour (cph) MSL 1 preconditioning: JESD22-A113-F Reflow temperature: at 260 +0/-5°C	231 (77 × 3 assembly lots per SAT)	Pass	Pass
Unbiased highly accelerated stress test: JESD22-A118 MSL 1 preconditioning: JESD22-A113-F Reflow temperature: 260 +0/-5°C	231 (77 × 3 assembly lots per SAT)	Pass	Pass
Biased highly accelerated stress test: JESD22-A110 130°C/85% RH and 96-hour duration Preconditioning: JESD22-A113 MSL 1, reflow temperature: at 260 +0/-5°C	231 (77 × 3 assembly lots per SAT)	Pass	TBD
High-temperature storage life: JESD22-A103-D Temperature 150°C, 1000 hours	231 (77 × 3 assembly lots per SAT)	Pass	Pass
Physical dimensions: JESD22-B100-A Package outline drawing: NT90-PK502-1	15 (15 × 1 assembly lot)	Pass	Pass
Solder ball shear: JESD22-B117-B (total samples from three assembly lots at each SAT)	45 (15 × 3 assembly lots)	Pass	Pass

7.2 Qualification sample description

Table 7-5 Device characteristics

Category	Definition
Device name	PM6125/PM4250
Package type	FOWNSP144
Package body size	4.81 mm × 4.81 mm × 0.57 mm
Solder ball composition	SAC405
Process	150 nm
Fab sites	■ F = F for TSMC ■ F = P for SMIC ■ F = J for Samsung
Assembly sites	■ A = E for ASE, Taiwan ■ A = Z for Amkor Technologies, Portugal
Solder ball pitch	0.40 mm

8 Revision history

Revision	Date	Description
A	November 2018	Initial release
B	February 2019	■ Table 2-6 Pin descriptions – general-purpose input/output functions: Updated the GPIO special functions ■ Section 4.2 Part marking: Added this section ■ Section 4.3 Device ordering information: Added this section ■ Section 4.4 Device moisture-sensitivity level: Added this section ■ Section 4.5 Thermal characteristics: Added this section ■ Chapter 5 Carrier, storage, and handling information: Added this chapter ■ Chapter 6 PCB mounting guidelines: Added this chapter ■ Section 7.2 Qualification sample description: Added this section
C	May 2019	■ Chapter 3 Electrical specifications: Added this chapter ■ Table 4-1 PM6125 device marking line definitions: Updated the fab source and assembly site code ■ Table 4-3 Device identification details: Updated the device identification details for CS samples ■ Table 7-1 Device characteristics: Updated the fab source and assembly site code
D	August 2019	■ Table 3-19 Analog multiplexer and scaling functions: Updated the source details of AMUX3_GPIO and AMUX4_GPIO ■ Table 3-20 VADC electrical specification: Updated the pull-up temperature coefficient test conditions and specification ■ Section 7.1 Reliability qualifications summary: TSMC and SMIC: Added this section
E	November 2019	■ Global: Updated the document for PM4250 ■ Table 2-4 Pin descriptions – general housekeeping functions: Updated the functional description for DVDD_BYP ■ Table 4-1 PM6125/PM4250 device marking line definitions: Added Samsung foundry ■ Table 4-2 Device identification code: Updated the device identification code for PM4250 ■ Table 4-3 Device identification details: Updated the device identification details for PM4250 ■ Table 4-4 Source configuration code: Updated the S code for Samsung foundry ■ Section 4.5 Thermal characteristics: Updated the thermal document references ■ Table 7-2 Package reliability results for SMIC: Updated the package reliability results ■ Table 7-3 Package reliability results for TSMC: Updated the package reliability results

Revision	Date	Description
F	December 2019	■ Table 2-7 Pin descriptions – general-purpose input/output functions for PM4250 on SM4250/SM6115 platform: Added this table ■ Table 3-8 PM4250 SMPS regulator summary: Added this table ■ Table 3-9 PM4250 linear voltage regulator summary: Added this table ■ Table 3-18 PM4250 power-on sequence for SM4250/SM6115: Added this table
G	May 2020	■ Table 2-3 Pin descriptions – output power management functions: Updated the functional description of VDD pins ■ Table 3-13 LDO regulator specifications: Added overall DC error specifications ■ Table 3-17 PM6125 power-on sequence for SM6125: Updated the footnote and option pin configuration. Added details for VREG_L9_LVS and VREF_LPDDR3 ■ Table 3-18 PM4250 power-on sequence for SM4250/SM6115: Updated the voltage details and the footnote ■ Section 3.6.2 FT-SMPS: Added FTS efficiency plots ■ Section 3.6.3 HF-SMPS: Added HFS efficiency plots ■ Section 3.9.1 PM6125/PM4250 OPTION hardware control: Added a note
H	September 2020	■ Table 1-1 PM6125/PM4250 features: Updated the pseudocapless LDOs ■ Table 2-4 Pin descriptions – general housekeeping functions: Updated the VREF_LPDDR3 functional description ■ Table 3-6 PM6125 SMPS regulator summary: □ Updated the specified voltage range for S6, S7, and S8 regulators □ Updated the rated current value for S7 regulator ■ Table 3-8 PM4250 SMPS regulator summary: Updated the rated current value for S7 regulator ■ Table 3-19 Power-on timing specifications: Updated the table
Revision I was omitted in accordance with QTI document conventions		
J	January 2021	■ Table 4-3 *Device identification details*: Updated the device identification details for PM6125 and PM4250 ■ Table 4-4 *Source configuration code*: Updated the source code for Samsung foundry
K	August 2022	■ Table 2-6 *Pin descriptions – general-purpose input/output functions for PM6125 on SM6125 platform*: Added the footnote ■ Table 2-7 *Pin descriptions – general-purpose input/output functions for PM4250 on SM4250/SM6115 platform*: □ Updated the functional description for GPIO_01 □ Added the footnote ■ Section 3.7 *GPIO specifications*: Updated the text to 'Eight out of nine GPIO ports are digital I/Os that can be programmed for a variety of configurations'.
L	January 2024	■ Table 3-5 *GPIO_MV digital I/O characteristics*: Added this table ■ Table 3-6 *GPIO_MV load current drive*: Added this table ■ Table 3-7 *GPIO_LV digital I/O characteristics*: Added this table ■ Table 3-8 *GPIO_LV load current drive*: Added this table ■ Table 7-1 *Silicon reliability results: SMIC, TSMC, and SEC*: Added this table ■ Table 7-4 *Package reliability results for SEC*: Added this table

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